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AfCFTA: Updated Analysis of Trade, Economic Outcomes, and Policy Implications

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Interactive infographic

The reader is invited to consult the EC data portal of agro-economic modelling DataM at https://datam.jrc.ec.europa.eu for more details of the modelling results.

The interactive infographic about this study is under the PANAP section at https://datam.jrc.ec.europa.eu/datam/area/PANAP

Direct link: https://datam.jrc.ec.europa.eu/datam/mashup/AFCFTA_FOLLOWUP/index.html

Contents

Abstract

This study provides a comprehensive and updated analysis of the potential impacts of the African Continental Free Trade Area (AfCFTA). Utilizing the MAGNET model, based on GTAP version 7 and database version 11C, this research incorporates new information such as actual tariff offers, revised Non-Tariff Measure (NTM) modelling, and updated baseline projections to offer a more accurate assessment than previous studies.

The simulations indicate positive economic outcomes for Africa, including increases in GDP, household disposable income, and intra-African trade. While tariff reductions alone have a modest impact, the reduction of NTMs is crucial for achieving substantial benefits, with spillover effects to extra-African trade further amplifying these gains. The study also highlights that the AfCFTA is expected to foster industrialization in Africa, with manufacturing output showing the most significant increases among sectors.

However, the analysis also points to regional disparities in economic benefits and potential challenges such as a decrease in tariff revenues for some countries. The findings underscore the importance of complementary policies, effective NTM reduction, and strategic management of public revenue streams to ensure equitable and sustainable growth across the continent. Future research is suggested to explore the long-term social and environmental impacts, as well as the institutional and regulatory frameworks necessary for the agreement's full implementation.

Introduction

Economic integration has a great potential to enhance the economic prospects of Africa. The continent is poorly integrated and depends on non-African trading partners. The extra-African trade is typically skewed towards lower value-added commodities. Consequently, Africa has had very limited opportunities to benefit from the economic growth of its trade partners, while at the same time it is exposed to highly volatile commodity prices. As a result, Africa has performed poorly in the process of catching up the economic development of richer countries. Better economic integration could alleviate the problem and help to foster thus far lacklustre industrialization (McMillan and Zeufack 2022).

A higher level of integration could be crucial for attracting more foreign investments to Africa. This is an increasingly challenging task in the current global context of rapid and capital-intensive technological change. For instance, maintaining adequate productivity improvements requires adopting new technologies of such factor input combinations that are increasingly at variance with African countries' factor abundance (Diao et al., 2025). Therefore, Africa faces ever higher challenges in adopting new technologies that would be essential for catching up with more developed countries. The African manufacturing sector increases mostly with small and informal firms while formal sector growth is modest when compared to other developing countries (McMillan and Zeufack, 2022). This development undermines the productivity growth and expansion of manufacturing exports necessary for reducing poverty in Africa. The authors propose that the African Continental Free Trade Agreement (AfCFTA) is one of the opportunities for changing this pattern. The agreement initiated the creation of the continent wide free-trade area (FTA) that specifically aims to deepen the economic integration between the African countries by reducing trade barriers and thus increasing intra-African trade.

The AfCFTA was established in 2018 when 44 of the 55 African Union (AU) member states signed the agreement. The trading under the AfCFTA was launched in practice in October 2021 by gradual reduction of intra-African tariff rates. The process is set to be completed by 2033 although the implementation has several issues pending. Significantly, the practicalities of the non-tariff measures (NTMs) reduction are not agreed upon even though several economic assessments have underlined its significance for achieving the full benefits of the integration.

In 2021, the Joint Research Centre (JRC) conducted a comprehensive study on the African Continental Free Trade Area (AfCFTA) for the African Union Commission Agriculture, Rural Development, Blue Economy, and Sustainable Environment (ARBE) Department (Simola et al., 2021). The study was based on the computable general equilibrium (CGE) model simulations with the MAGNET model (Woltjer et al. 2014). The study focused on the agri-food sectors, with a subsequent article examining the effects of implementing AfCFTA on food security (Simola et al., 2022). All results of the study are available in a data dashboard and have served the AU and AfCFTA negotiating parties. Four years after, we are updating the AfCFTA impact assessment on individual member countries by taking advantage of new information and methodological advances. These developments were relatively simple to incorporate in the present study and thus making timelier and accurate results available become a reasonable undertaking. The previous study also suffered from its proximity to the Covid-19 crisis and its effects on the African economies were not considered. The new baseline projections improve on that.

In addition, we can address more precisely a central uncertainty in the analysis. A large share of the benefits of the AfCFTA are estimated to be due to the reduction of costs associated with the non-tariff measures (NTMs), but there is considerable uncertainty over how much these costs can be expected to be reduced. There is also a potential for the NTM effects to spillover to extra-African trade flows. For example, by harmonizing an NTM between African countries, the new adopted methods can reduce the trade barriers also in extra-African trade flows. One example could be a simplified technical standard that would also reduce the costs between African and non-African countries. However, there is very little evidence on such spillovers. Therefore, in this study we vary the assumption of the spillover and see how that would affect the conclusion of the analysis.

This report summarizes the analysis on the AfCFTA with focus on the uncertainty around the NTMs. In addition, the report benefits from several points to update the previous approach, and the following changes have been included in the analysis:

1. Tariff offers: The initial study relied on speculative scenarios as actual tariff offers were not published. This study is able to base the scenarios on the actual tariff offers that have been tabled to this date, providing a more accurate basis for the analysis.[1] In addition, we include the original tariff revenue maximization scenario for comparison, and a hypothetical scenario fully eliminating tariffs to illustrate the potential gains from further liberalisation. The tariff shocks are also be based on an updated MacMaps tariff data of year 2017.
2. Non-Tariff Measures (NTMs): In the initial study, the reduction rate of NTMs in the intra-African trade was assumed 50% as a reasonable approximation. At the same time, we assumed a 25% reduction in the extra-African trade as a spillover effect. This study maintains the 50% reduction of the NTMs in the intra-African trade but also calculate a more modest scenario of 25% reduction. It also explores three levels of external trade spillover effects: 0%, 10% and 20%. This provides a clearer indication of how much African countries could gain by aligning their NTMs with those of their non-African trade partners.
3. In addition, we have revised the modelling of the NTMs, and we apply a 50/50 allocation of NTMs between exporters and importers as efficiency shocks as suggested by Walmsley et al. (2021). By using the efficiency shocks as the basis for NTM reductions, we eliminate the need for the potentially distorting Altertax tariff adjustment technique (Malcolm 1998). The NTM shocks are based on an updated data of year 2017, which is also regionally more detailed.
4. GTAP model and database: The initial study was based on GTAP model version 6 and GTAP database version 10. The updated simulations are based on GTAP version 7 (Corong et al. 2017) and GTAP database version 11C (Aguiar et al. 2022). The new model version has several new features including the investments that are sourced of several commodities rather than modelled as a single capital commodity[2], and a non-diagonal sourcing matrix that allows modelling of by-products. The update of the database is particularly significant as it incorporates the shift in the base year from 2014 to 2017 and includes 19 new regions, eleven of which are in Africa[3]. Thus, the new analysis can yield more regional detail, to deepen understanding of the AfCFTA's effects on various African regions. Moreover, the new database aligns with the FAO's agricultural data, providing a more realistic representation of agri-food sectors and food security.
5. The baseline projections: The initial study had a baseline constructed of two parts: initial historic baseline based on the IMF's World Economic Outlook (WEO) for a mid-term economic forecast, and longer-term projection based on the Shared Socioeconomic Pathways (SSP) version 2.0 on national economic and population projections from 2018. Both baseline components are updated in this study. First, the historic periods from the base year 2017 to 2029, is based on the October 2024 WEO release. This data considers the recent events like the Covid-19, Russia's invasion in Ukraine, Israel-Palestine conflict and how they affect mid-term economic projections. In general, the economic prospects of the African countries are revised downwards due to these events. Second, the long-term projection of the updated study is based on the SSP version 3.0 from 2024 that markedly differs in its economic and population trajectories of the African countries. Along with the

[1] In cases where tariff offers are still missing or only partial, we assume that the tariff reductions is chosen based the assumption of maximizing the remaining tariff revenues.

[2] Commodity CGDS in the earlier GTAP model versions.

[3] The new regions are Algeria, Central African Republic, Chad, Comoros, Congo, Democratic Republic of Congo, Equatorial Guinea, Eswatini, Gabon, Mali, Niger and Sudan. In addition, the IO tables of Botswana, Cameroon, Côte d'Ivoire, Mozambique, South Africa and Zimbabwe are updated.

population projections, the Wittgenstein educational attainment projections that define the skilled and unskilled labour demand was updated to the new version 3.0 as well. We run the SSP baseline until 2040, therefore adding a five year period after the finalization of the AfCFTA in 2035.

6. The MAGNET model: The dynamics of the MAGNET model have been significantly enhanced by a new wealth module that allows more flexible and realistic allocation of capital among industries. In the new version the capital demand is modelled independently of the GDP growth to which it was tied in the previous study. The new module also addresses an inconsistency in time step lengths in the earlier MAGNET versions that has biased the capital accumulation process in the model. It also includes the flows of foreign assets and liabilities and therefore takes into account the role foreign direct investments (FDI). The newest GTAP database version 11C used in the study includes updated data source for capital stocks and depreciation thus further enhancing the simulation of capital accumulation.

By implementing these changes, this follow-up study enhances the understanding of the AfCFTA's impact, utilizing more accurate and up-to-date data, and considering various aspects of the agri-food sectors and food security. This follow-up study also covers more regional detail than the preceding one. It also addresses the main uncertainties of the NTM reductions by including more varied scenarios on both intra- and extra-African NTMs.

Method

We apply the MAGNET model in the analysis (Woltjer et al. 2014) as in the earlier study, although the method has gone through the several enhancements described above. MAGNET is a global recursive dynamic CGE model based on the GTAP model version 7 (Corong et al. 2017) and database version 11C (Aguiar et al. 2022). To analyse the AfCFTA with as much regional detail as feasible, we applied an aggregation of 58 regions, which includes all 43 African regions in the GTAP database as individual regions. Of the 43 African regions five are composite regions and the rest individual countries (see Annex 1 for the regions and their mapping to the GTAP regions).

Baseline

A dynamic CGE analysis is based on economic trajectories that represent a plausible idea of the economic development on a timeframe relevant for the study. This baseline covers the years 2017 to 2040. The starting point 2017 is determined by the latest GTAP database base year, and it allows to use the most up-to-date consistent global input-output dataset. The economic development between 2017 and 2040 is a combination of two data sets: the IMF's World Economic Outlook (WEO; 2017-2029) (IMF 2024) and the Shared Socioeconomic Pathways (SSP; 2030-2040) (IIASA 2024).[4] The former is a mix of actual data and mid-term projections, while the latter is a long-term economic projection running until 2100. The combination of these two datasets achieves three things: 1) a more realistic starting point in the first policy year 2022 that is mostly based on actual economic data of WEO, 2) mid-term projections that take into account the latest assessment of global economic development that include the COVID-19 pandemic, the various on-going military conflicts and predicted subsequent adjustment processes, and 3) final year of the analysis being consistent with the widely used long-term economic projections.

We apply the GDP and population trajectories from the abovementioned sources directly in our baseline. In addition, we take advantage of the human capital data (KC et al. 2024) of the Wittgenstein Centre to determine the shares of the working age population and shares of skilled and unskilled labour. By applying these two shares we can construct labour demand projections for the two skill categories. The resulting changes in capital and land use, investments, private consumption and trade flows are results of the model process where the economies adjust to equilibrium prices and quantities. In addition to these macro shocks, the MAGNET applies exogenous agricultural land productivity (yield), and animal feed efficiency shocks based on the IMAGE model projections.

Exogenous baseline

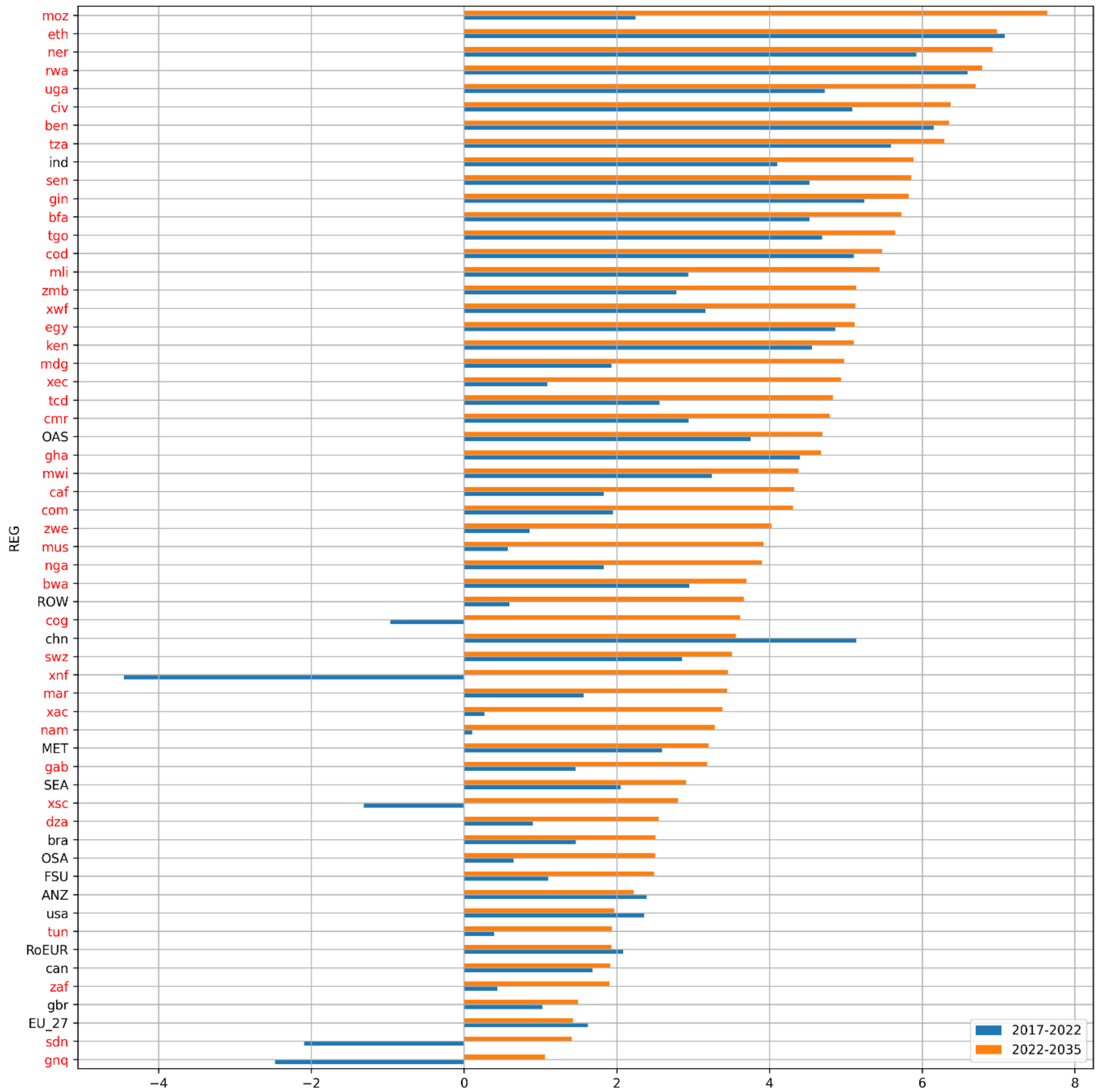
A dynamic CGE model baseline assumes some exogenous trajectories like the ones described above. This subsection describes the exogenous part of the baseline. Figure 1 depicts the yearly GDP growth rates for each model region in the baseline for two time periods: before the policy changes (2017-2022) and during the implementation of the AfCFTA (2022-2035). Many African countries are found on the top of the ranking, and that well illustrates how the catching-up effect is assumed to benefit many of the African countries in the baseline. In the top ten countries with highest growth rates in 2022-2035, only one (India) is not an African one. At the same time there are four African countries (Equatorial Guinea, Sudan, South Africa and Tunisia) in the bottom ten regions with the slowest growth. This illustrates the divergence in the predicted fortunes within African countries in the mid-term projections.

Figure 2 depicts the development of the population growth, and showing that a large majority of the regions with the highest population growth rates are African. The continent has the highest overall population growth, and it is predicted to increase from the current 1.5 billion to 2.5 billion by 2050, thus making more than every fourth person in the planet African (Stanley 2023). In many

[4] The WEO data is incomplete for some country-year observations, and in those cases the values from the SSP data was used.

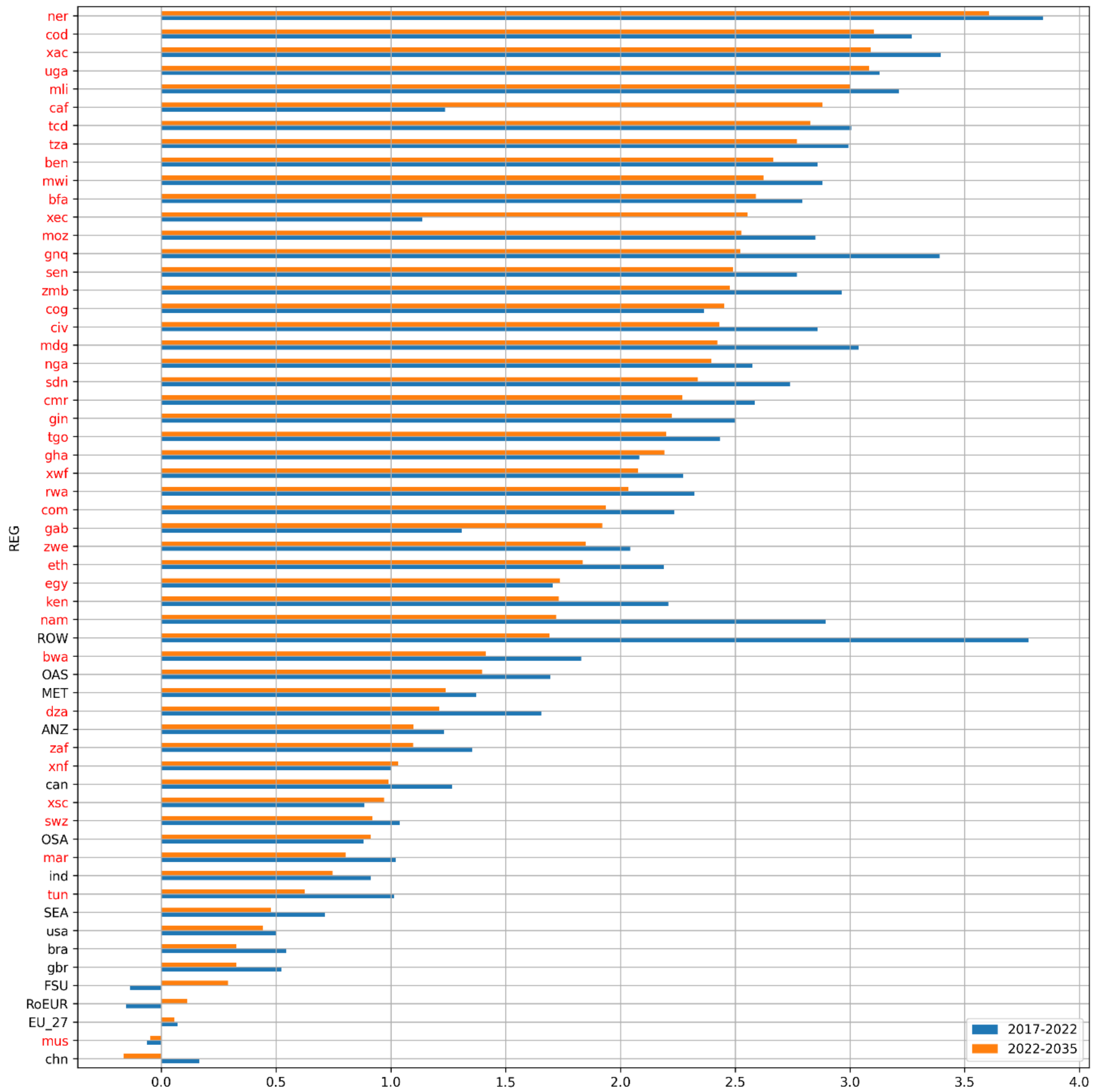
cases high population growth consumes a large share of the economic growth, and consequently the per capita growth rates of GDP show even more striking divergence in Africa (Figure 3). In the top ten there are two non-African regions (India and China), whereas in the bottom ten we find seven African regions (Equatorial Guinea, Sudan, Rest of South and Central Africa, South Africa, Congo, Gabon and Tunisia). The divergence in per capita GDP projections highlight the prediction that poverty reduction remains a challenge for several African regions on short and medium term.

Figure 1. GDP growth rates on the baseline (yearly percentage change within the period)



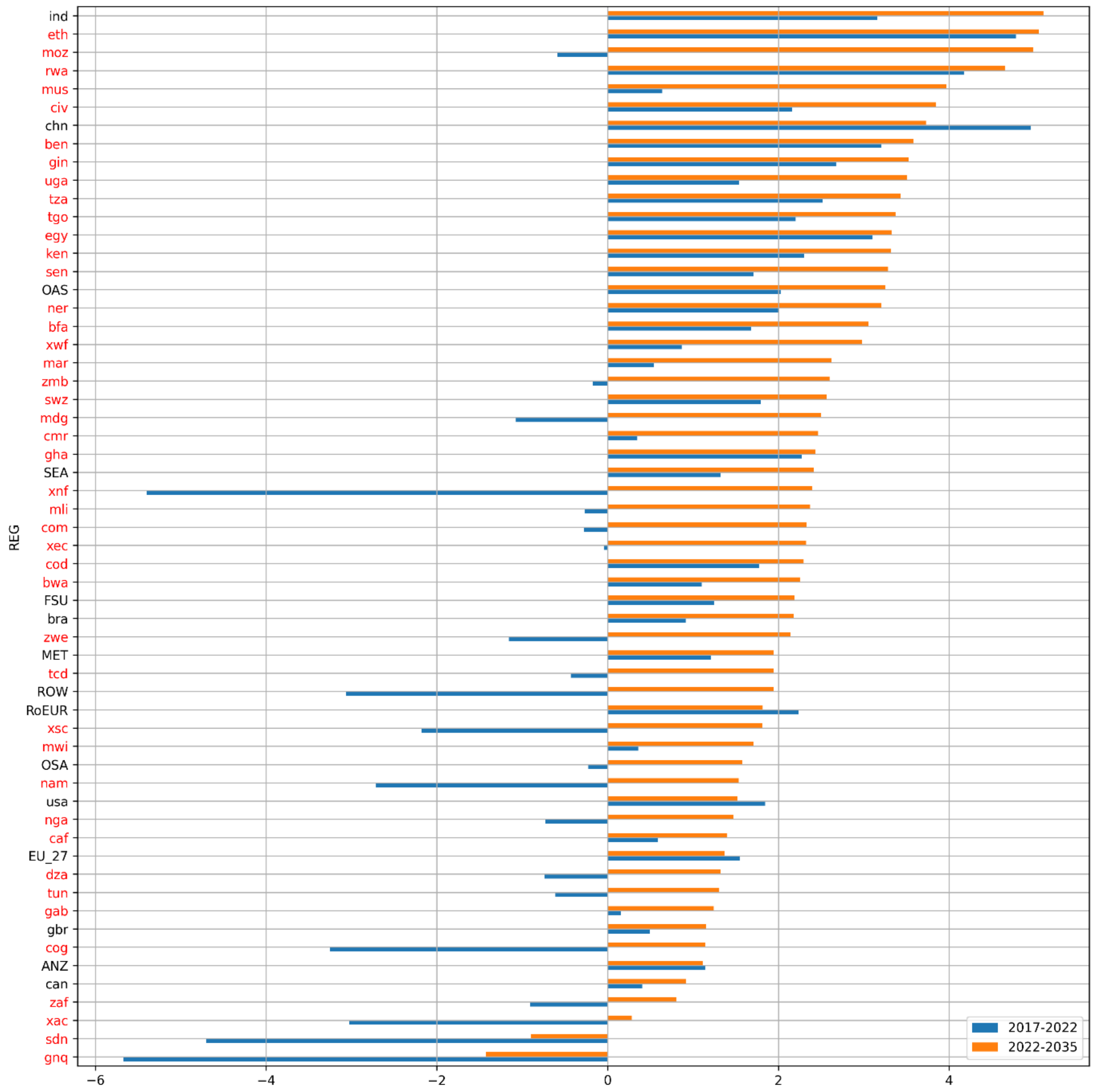
Source: the MAGNET baseline derived from IMF WEO and SSP projections; Note: The African regions are marked in red.

Figure 2. Population growth rates on the baseline (yearly percentage change within the period)



Source: the MAGNET baseline derived from IMF WEO and SSP projections; Note: The African regions are marked in red.

Figure 3. GDP per capita growth rates on the baseline (yearly percentage change within the period)



Source: the MAGNET baseline derived from IMF WEO and SSP projections; Note: The African regions are marked in red.

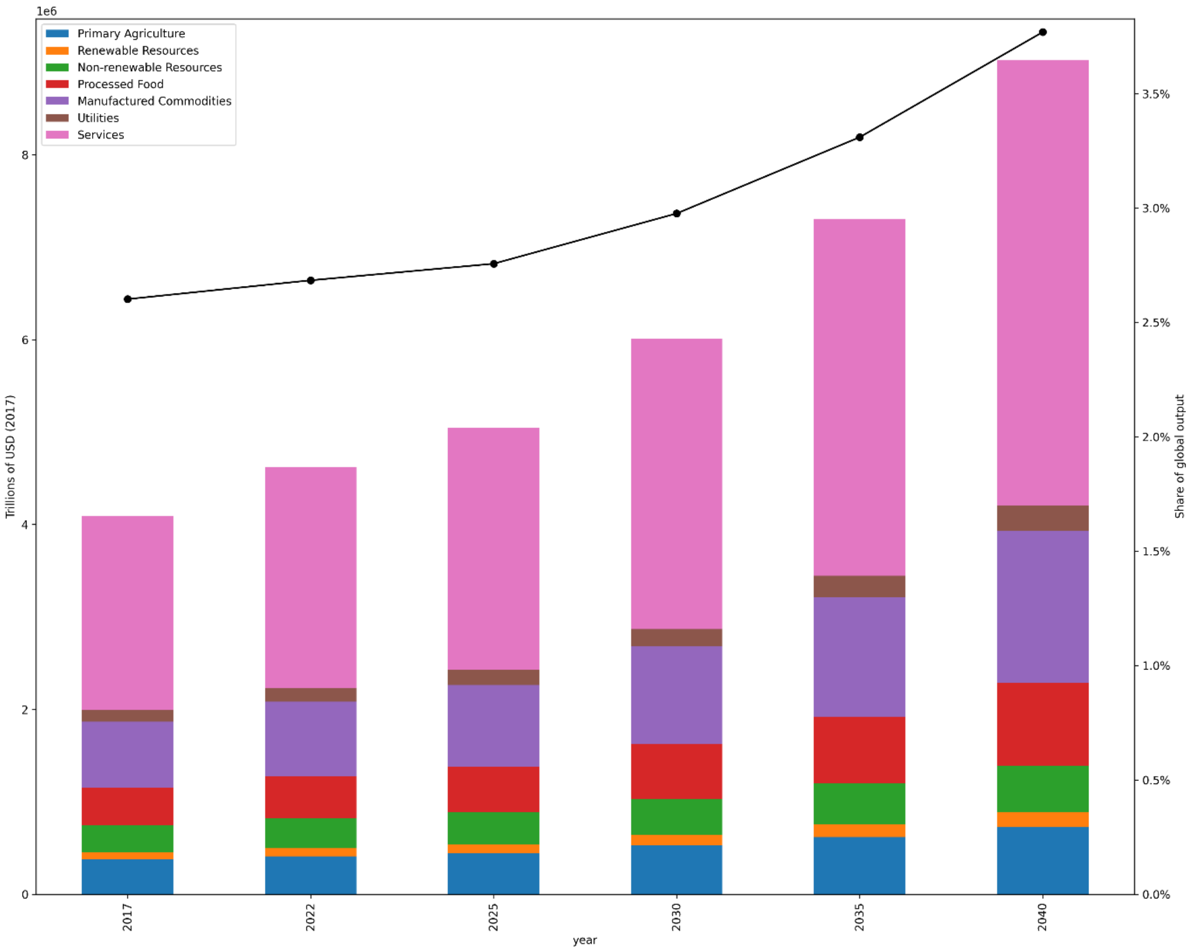
The preceding figures also show the corresponding growth rates for the pre-policy simulation period 2017-2022 that is mostly based on actual macro-economic WEO data. The period was exceptionally difficult to predict with several impactful events such as COVID-19 and increased occurrence of military conflicts that was highlighted by large fluctuations. With only a handful of exceptions, all the regions with negative GDP per capita growth rates in 2017-2022 are African, further illustrating the economic vulnerability of the region for COVID-19 and other shocks.

Endogenous baseline

In addition to exogenously determined macroeconomic trends, the MAGNET model endogenously determines several baseline developments, including output, final consumption and trade flows at commodity levels. This section briefly describes the development of these flows for Africa in total.

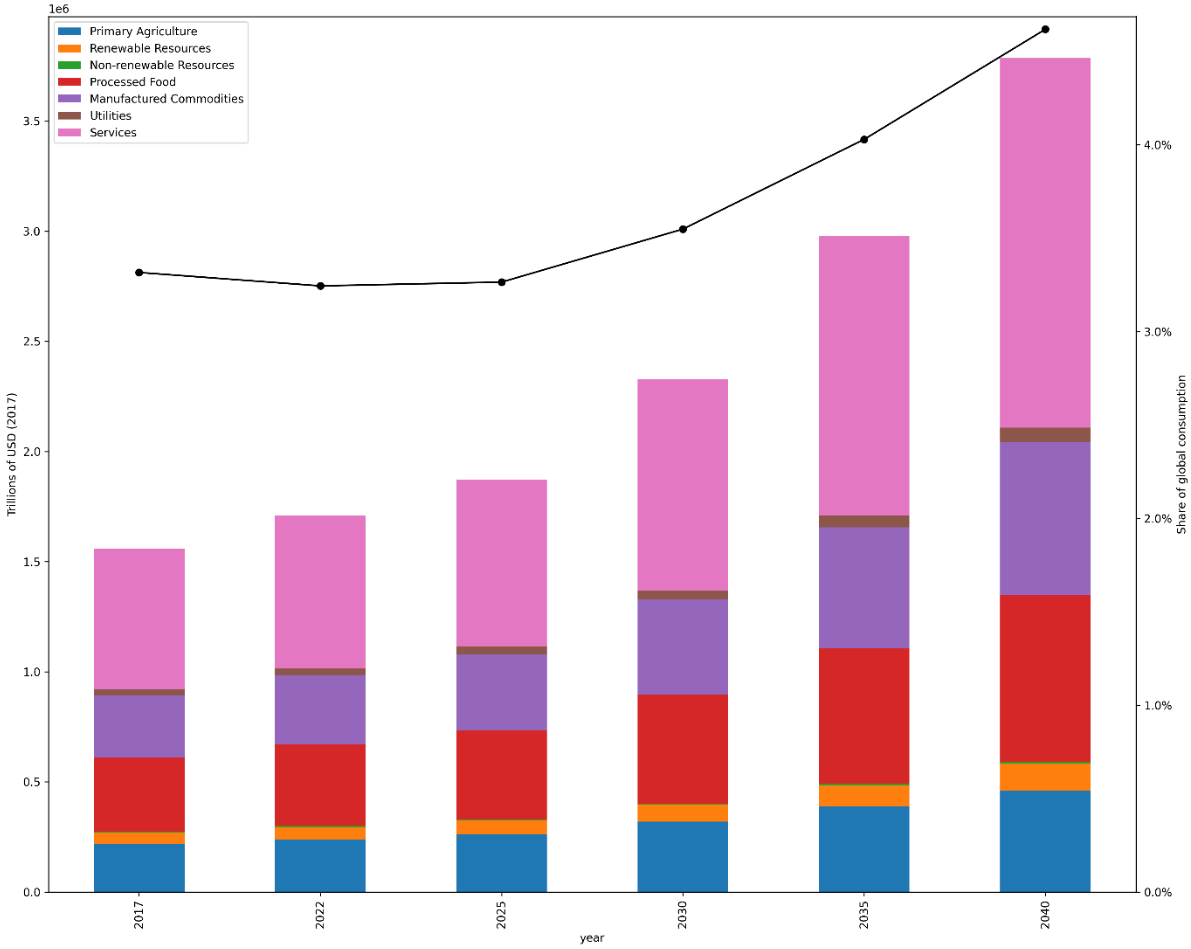
As the GDP grows in the baseline, its commodity composition also changes (Figure 4). Output of all the commodity categories increase in absolute terms, but especially services increase in their share. The same happens to manufacturing commodities, but in smaller extent. Along with a development to more service-oriented economies, the African share of total global economy increases, reaching 3.8% in 2040. The same overall pattern holds for the household consumption (Figure 5).

Figure 4. Total output by commodity groups in Africa (in real terms; trillions of 2017 USD)



Source: the MAGNET baseline calculations.

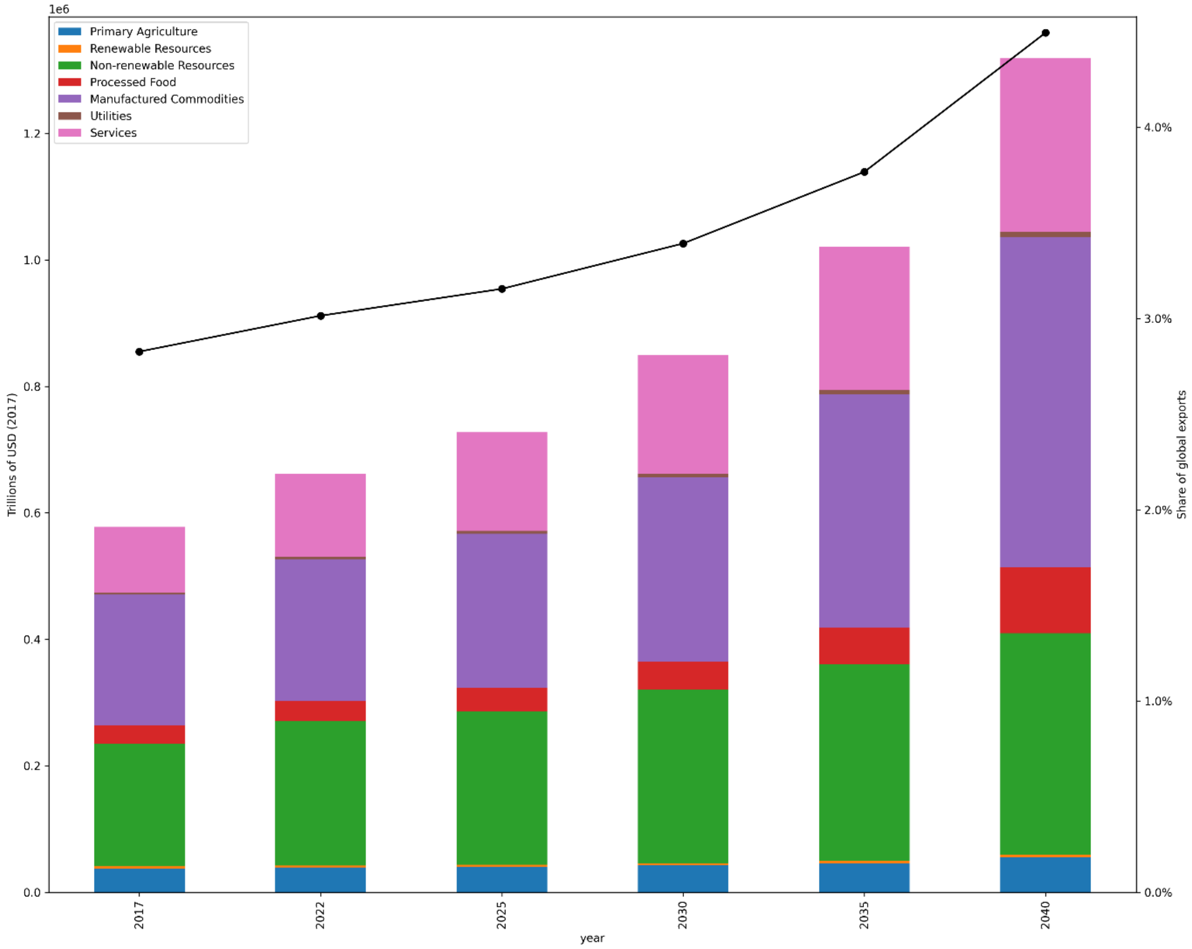
Figure 5. Total private consumption by commodity groups in Africa (in real terms; trillions of 2017 USD)



Source: the MAGNET baseline calculations.

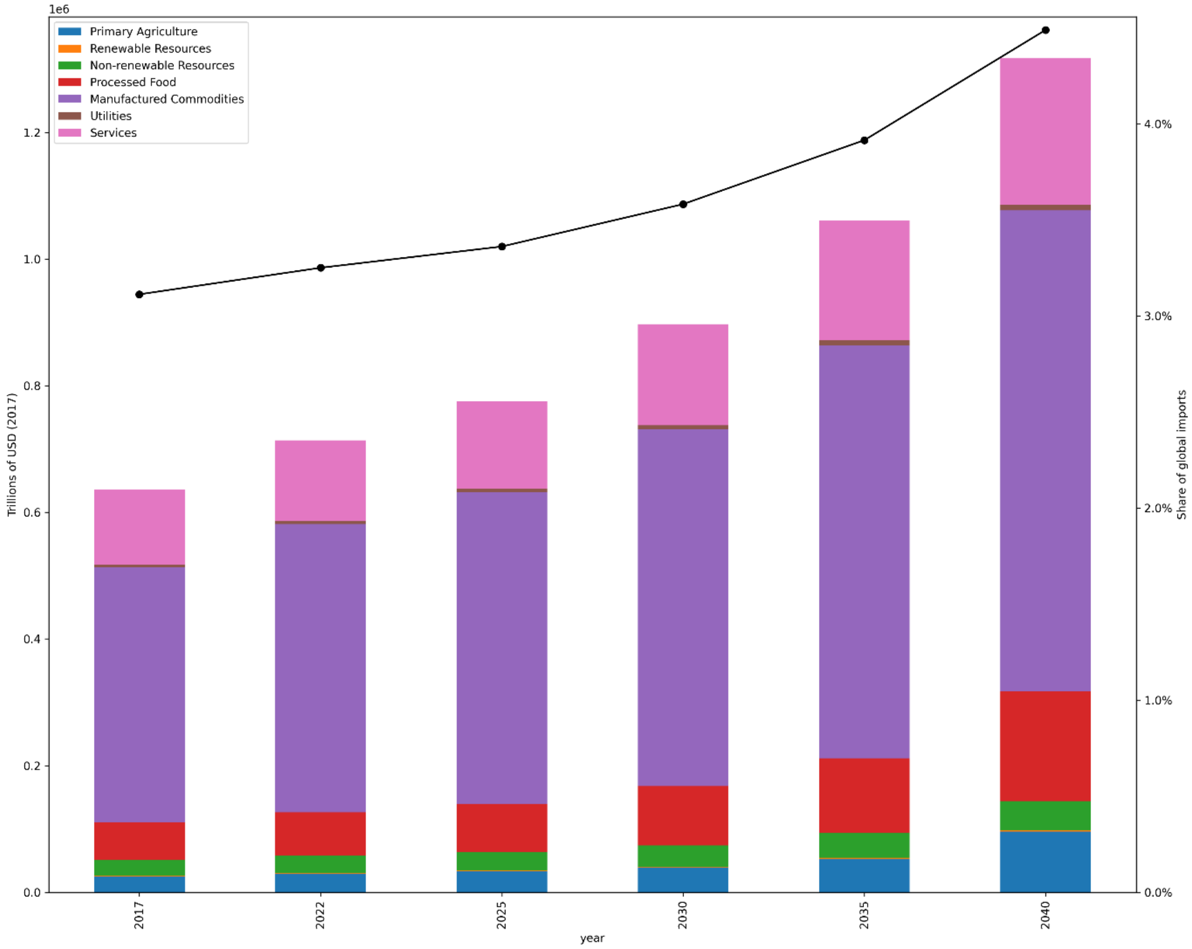
The trade volumes of the African economies also increase in the baseline. The figures illustrate that African countries mostly export natural resources and other low value-added commodities (Figure 6), while importing manufactured commodities (Figure 7). The continent is also a net importer of food commodities (Figure 7).

Figure 6. Total export flows by commodity groups in Africa (in real terms; trillions of 2017 USD)



Source: the MAGNET baseline calculations.

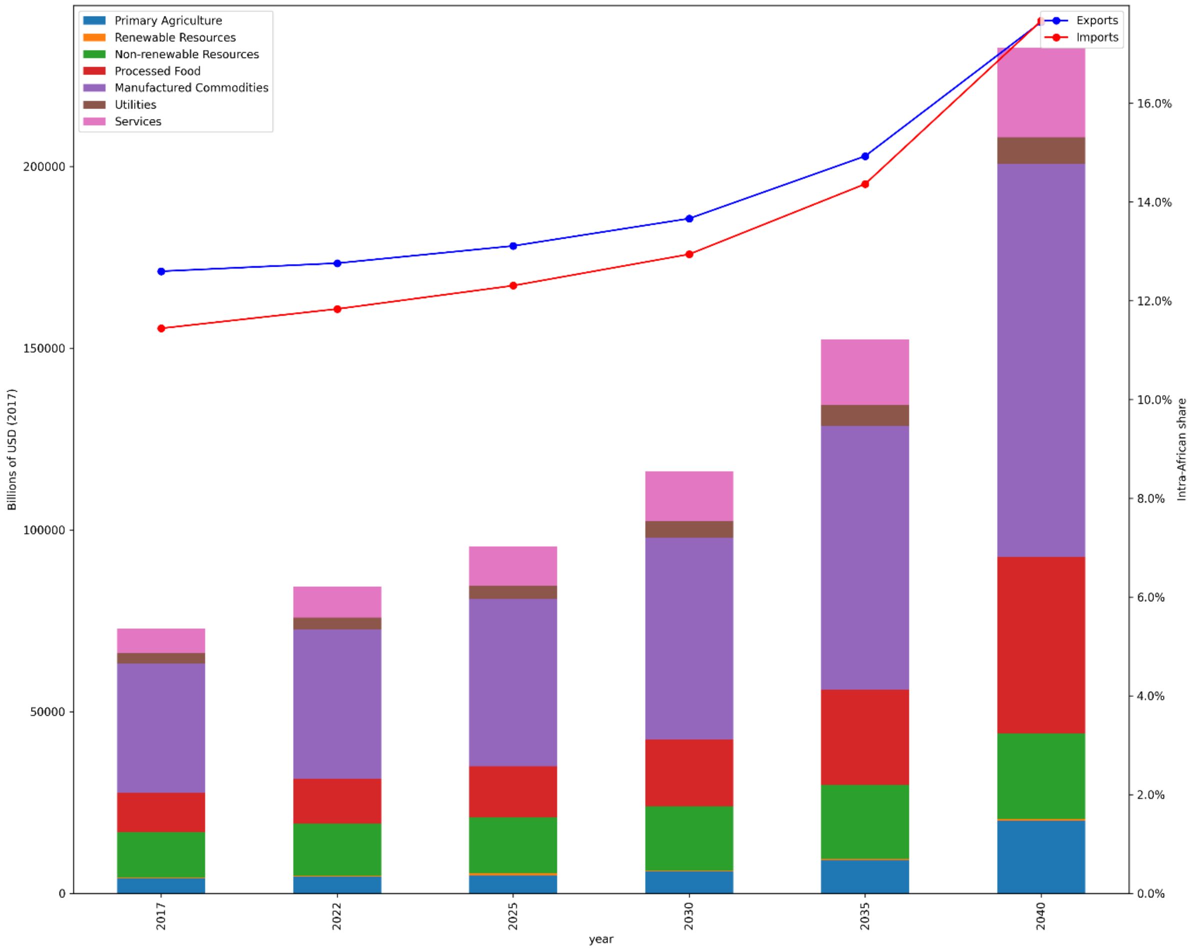
Figure 7. Total import flows by commodity groups in Africa (in real terms; trillions of 2017 USD)



Source: the MAGNET baseline calculations.

Africa also trades less within Africa than with non-African partners. Along with increasing incomes in Africa, the share of intra-African trade increases in our baseline, while the implementation of the AfCFTA is poised to contribute to further increasing the share. The intra-African trade flows (the imports equalling the exports) increase in absolute numbers and with respect to total trade volumes (Figure 8). The largest share of intra-African trade flows is composed of manufactured commodities.

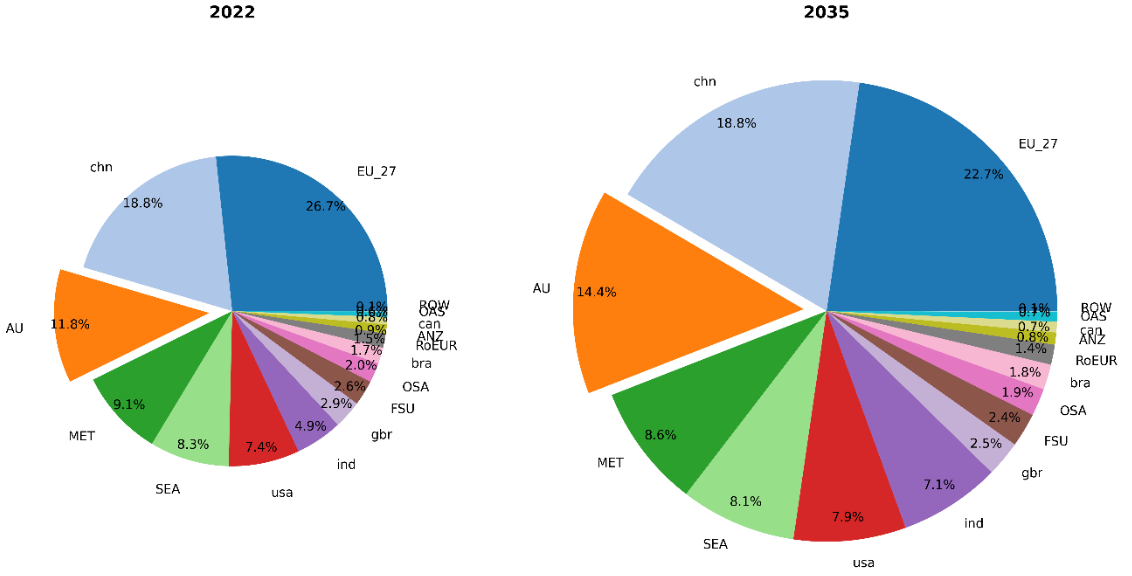
Figure 8. Intra-African trade flows by commodity groups in Africa (in real terms; billions of 2017 USD)



Source: the MAGNET baseline calculations.

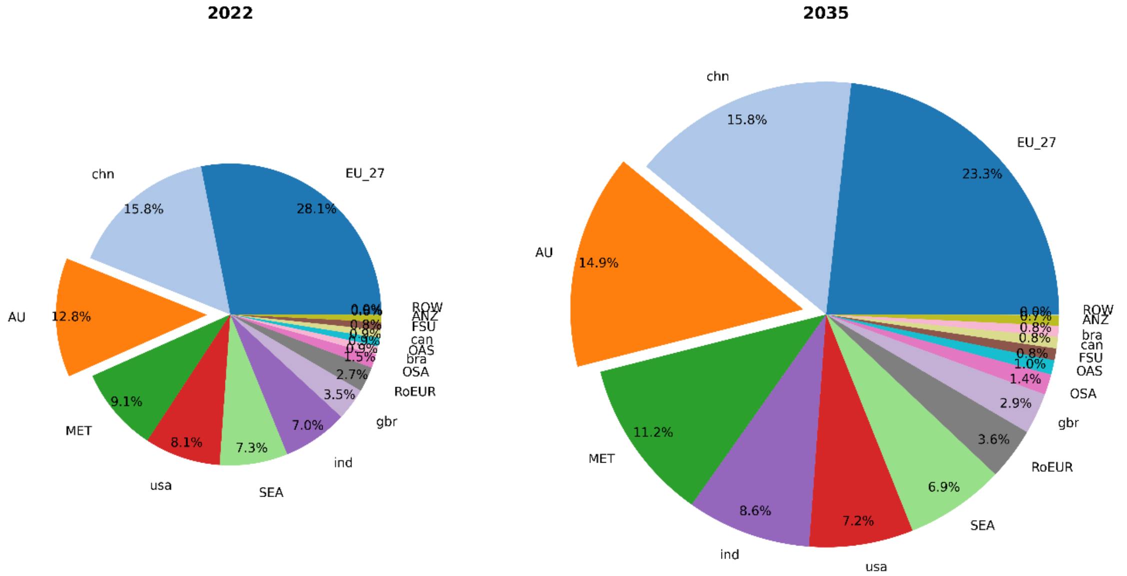
The regional composition of trade also evolves endogenously along the baseline. Figure 9 and Figure 10 illustrate the export and import compositions, respectively, at both the start (2022) and end (2035) of our simulation period. Notably, the share of the intra-African trade increases, confirming previous observations. While the EU's share is projected to decrease by 2035, it is expected to remain the most significant trading partner. Other major non-African trading partners, including China (chn), the Middle East (MET), Southeast Asia (SEA) and the USA, maintain relatively stable shares and rankings throughout the period. Another significant development is the growing importance of India, which sees an increase in its share of both imports and exports.

Figure 9. African export shares by destination (the start and end of the baseline)



Source: the MAGNET baseline calculations.

Figure 10. African import shares by source (the start and end of the baseline)



Source: the MAGNET baseline calculations.

Policy scenarios

We model the implementation of the AfCFTA with respect to changes in intra-African tariff rates and changes in NTMs. We apply three scenarios for tariff cuts: 1) maximization of tariff revenues (maxTR), 2) current offers (offer), and 3) full liberalization (zeroT). The first two are based on the tariff offer analysis by Boysen (2024) and they both comply with the AfCFTA tariff liberalisation modalities (Table 1). The *maxTR* scenario assumes that countries choose tariff lines to be sensitive or excluded from liberalisation, in accordance with the modalities, such that the tariff revenue is as high as possible at any point in time. The most realistic scenario is the *offer* scenario which is based on the available data on the actual offers and completed by the procedure described in Boysen (2024). Finally, the zeroT scenario assumes full, linear elimination of all the intra-African tariffs. This full liberalization scenario is purely hypothetical, and it serves as a measure of how much more could be achieved by eliminating the tariffs between African countries.

Table 1. Tariff line categories according to the AfCFTA modalities

Category	Label	Rule	To be eliminated over	
			Non-LDCs	**LDCs**
A	Non-sensitive	Tariffs on at least 90% of tariff lines to be eliminated.	5 years	10 years
B	Sensitive	No more than 7% of tariff lines may be declared sensitive. The liberalisation start may be delayed to year 6.	10 years	13 years
C	Excluded	No more than 3% of tariff lines may be exempted from liberalisation. The total value of these imports must not exceed 10% of the value of all imports from within the AU.		

Source: Boysen (2024).

In addition to the tariff cuts, the AfCFTA aims for reducing the NTMs. Specifically, the aim is to eliminate the non-tariff barriers (NTBs), the trade distorting part of the NTMs. Several previous studies (e.g. Jensen and Sandrey 2015, World Bank 2020 and 2022, Simola et al. 2021 and 2022) have shown that a large share of the economic benefits of the AfCFTA are due to the NTM reductions. There is however a considerable uncertainty surrounding the magnitude of the NTMs as they can only be indirectly measured, and there is very little evidence indicating by what percentage the trade-reducing effect of the NTMs could be reduced through the AfCFTA. In this analysis we make two assumptions on the NTM reductions due to the AfCFTA: 25% and 50%. Many studies have assumed a 50% reduction (e.g. World Bank 2020 and 2022, Simola et al. 2021 and 2022), although there is little empirical support for the feasibility of a reduction of that size. Therefore, our study adds a sensitivity check for the NTM reduction. The reduction happens linearly in 2022-2035 so that the levels of the NTMs in 2035 are reduced to 50% or 75% of their initial levels, depending on the scenario. We allocate the reduction evenly to exporters and importers, following the import-augmenting and export cost[5] approaches by Walmsley & Strutt (2021).

[5] The import-augmenting method is also known as the 'iceberg' method and it targets the GTAP variable *ams*. The export cost method is the export side variation of the 'iceberg' method and targets a new variable *axs*.

Some earlier studies (e.g. World Bank 2020 and 2022, Simola et al. 2021 and 2022) have assumed that the intra-African NTM reductions could have spillover effects on the trade flows between African and non-African countries. There is no actual evidence of such spillover effects and inclusion of them in the analysis should be done with caution. Therefore, this study varies the basic intra-African NTM reduction assumption with three additional assumptions on the extra-African spillover effects: 1) no spillover, 2) 10% spillover and 3) 20% spillover to trade flows between African and non-African countries (Table 2). This wide array of assumptions on NTMs and their spillover effects also serves as sensitivity check of the results for various assumptions. In addition to being a sensitivity check, the various spillover effects can shed light on the effects on aligning the NTMs with the extra-African trading partners.

Table 2. Simulation scenarios

		NTM reductions	
scenario	tariff cuts	intra-African trade flows	extra-African spillover
base	0	0	0
tar	maxTR, offer or zeroT	0	0
tar_NTM25_ea00	maxTR, offer or zeroT	25%	0
tar_NTM25_ea10	maxTR, offer or zeroT	25%	10%
tar_NTM25_ea20	maxTR, offer or zeroT	25%	20%
tar_NTM50_ea00	maxTR, offer or zeroT	50%	0
tar_NTM50_ea10	maxTR, offer or zeroT	50%	10%
tar_NTM50_ea20	maxTR, offer or zeroT	50%	20%

Source: authors' own elaboration.

Note: Each policy scenario is calculated separately with every tariff reduction strategy.

Results

This section presents the outcomes of the policy simulations as percentage deviations from the baseline in year 2035 when the AfCFTA is fully implemented. Thus, for example, a 0.2% deviation in GDP means that in the policy simulation, the GDP outcome is 0.2% higher than it would have been in the baseline in the year 2035.

Impacts on the AU level

Figure 11 collects the results at the AU level for the GDP and its absorption components. The GDP effect is modest, especially considering the tariff cuts only. The tariff reduction strategy affects the GDP minimally, but it has more diverse effects on the absorption components. Both investments and private consumption increase too, the latter slightly more than the overall GDP. The investments increase markedly less. At the same time, the public spending is affected positively only when the NTMs are reduced with a sufficient extra-African spillover effect. This is a direct consequence of decreasing tariff revenues that affects most directly the scenarios with low levels of NTM effects. The strategy for maximizing the remaining tariff revenues yields the highest increases in public spending, whereas the current offers yield the highest investment levels. Both NTM reductions and NTM spillovers generate higher results for all the variables. In addition, the effect of the small incremental spillover increases (0% -> 10% -> 20%) produces higher GDP effects than the much larger NTM reductions (0% -> 25% -> 50%). This reflects the still large share of non-African trade flows that are affected by the spillovers, but not by the NTM reductions.[6]

Figure 12 summarizes the effects on the GDP decomposed at the income side. The figure includes also the extraction of natural resources that is a significant source of factor income in many regions. The aggregate factor income increases more than the GDP. The overall tax income decreases except in the cases where we assume spillover effects. Of the natural resources, the extraction of coal decreases in all scenarios, while crude oil and natural gas uses increase. The differences between the scenarios are relatively small for the natural resource use although the coal use decreases the most in the full liberalization strategy. Thus, the natural resources will remain an important part of the African income sources and increase in absolute terms although their share in overall factor income will decrease.

Figure 13 summarizes the effects on the real[7] changes in primary factor rental rates. As a general pattern, capital and labour categories' rents increase, while the land rental rates have a more mixed pattern: they increase in full liberalization scenarios and higher NTM and spillover changes. Capital rents increase more than labour rents, which is in line with the AU economy moving to a more capital-intensive production and higher levels of aggregate investments. On the other hand, the unskilled labour wages increase more than skilled labour wages, which means that the industries that employ more unskilled labour increase their activities more. The ranking of the strategies is the same throughout: full liberalization increases the rents the most, while the maximization of the tariff revenues has the smallest effect. The figure also includes the consumer price index (CPI) to illustrate the effects on the purchasing power parity (PPP): although the AfCFTA has inflationary effects, the wages increase more than compensate the price increases, allowing for a higher level of consumption.

Figure 14 summarizes the effects on the trade flows. Both total imports and exports increase in all scenarios. Only a small additional increase comes from NTM reductions. Imports increase slightly more in tariff revenue maximization and full liberalization strategies, and thus contributing to a small trade balance deterioration, which is ameliorated with higher spillover effects. Intra-African trade[8] increases more, an expected and desired result of lower trade barriers within the African

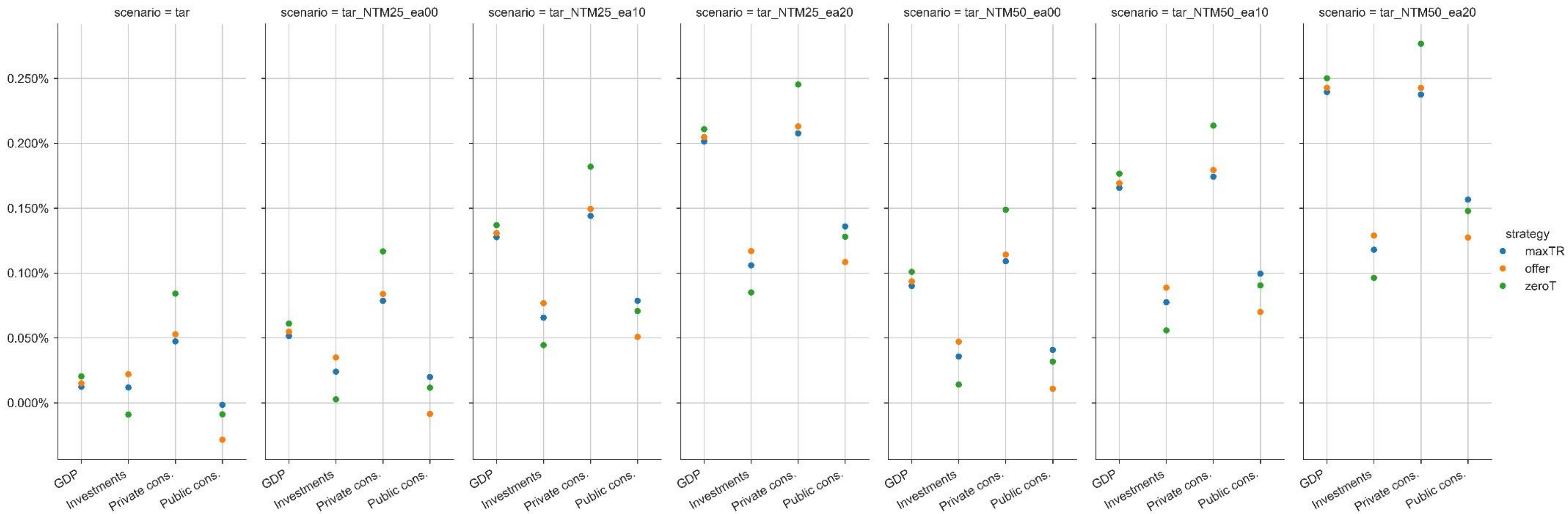
[6] This should also highlight the uncertainty in the spillover assumption and the need to conduct a sensitivity analysis with respect to it if applied.

[7] Adjusted with consumer price index (CPI).

[8] Intra-African imports and exports are equal by definition and are presented by a single column in the figure.

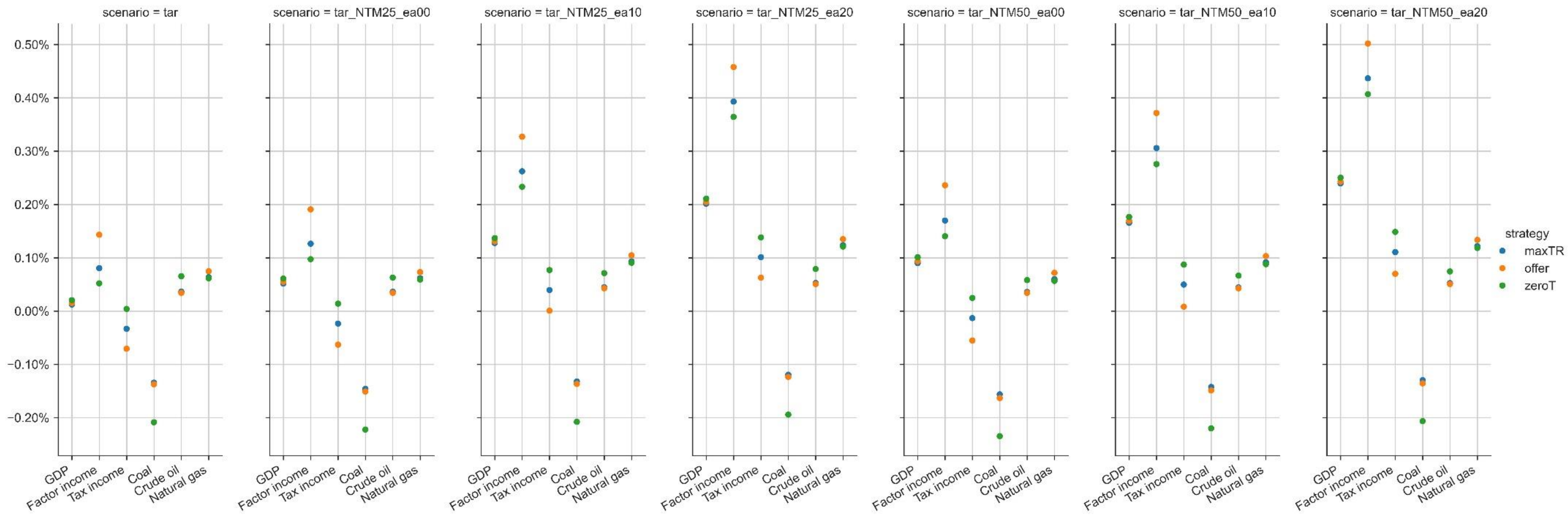
continent. Full liberalization achieves the highest increase, while the tariff maximization scenario the lowest. NTM reductions have a more tangible effect on the intra-African trade flows than the total trade flows. The trade flows with non-African countries decrease slightly in almost all cases. The spillover effects slightly decrease the effects on the intra-African flows, while reducing the reductions on the extra-African trade flows.

Figure 11. Economic impacts at the AU level – GDP and absorption



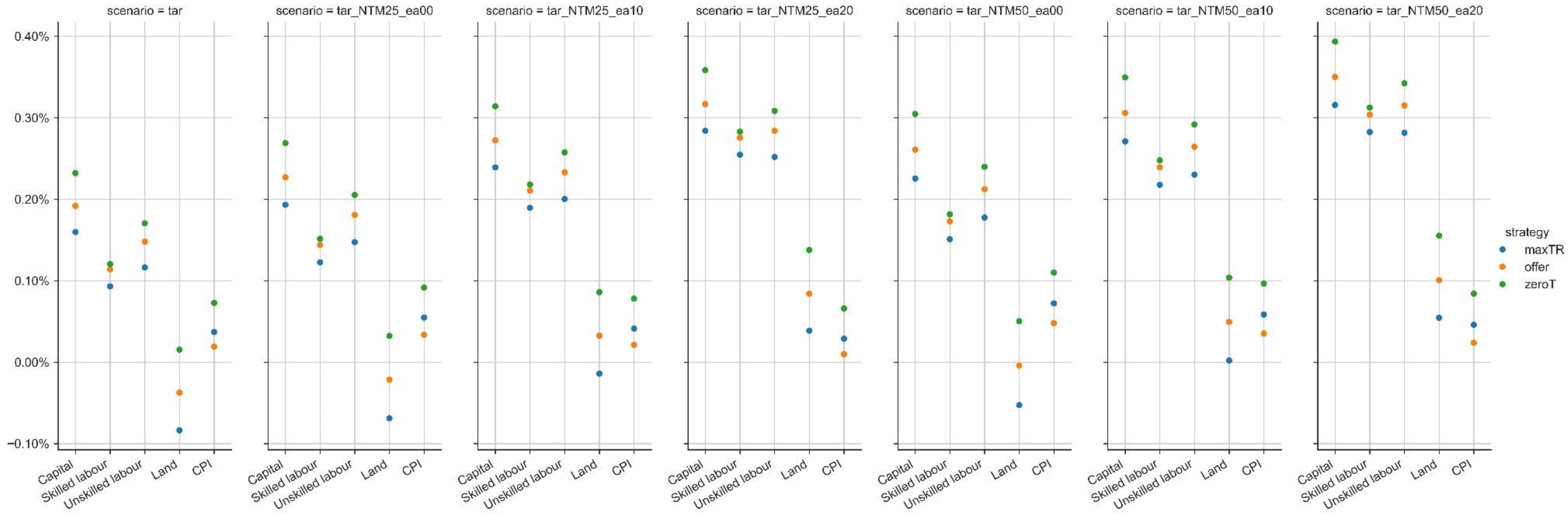
Source: the MAGNET calculations.

Figure 12. Economic effects at the AU level - GDP income side decomposition



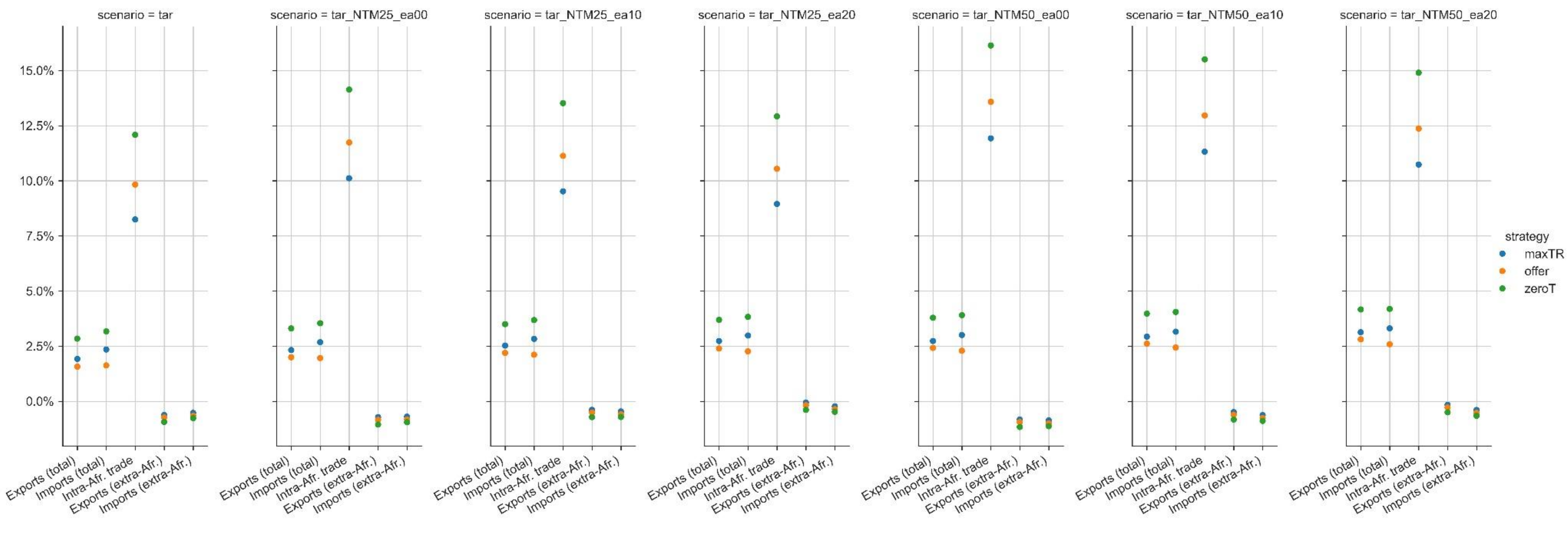
Source: the MAGNET calculations.

Figure 13. Economic effects at the AU level - factor rental rates (CPI adjusted)



Source: the MAGNET calculations.

Figure 14. Economic effects at the AU level - trade flows

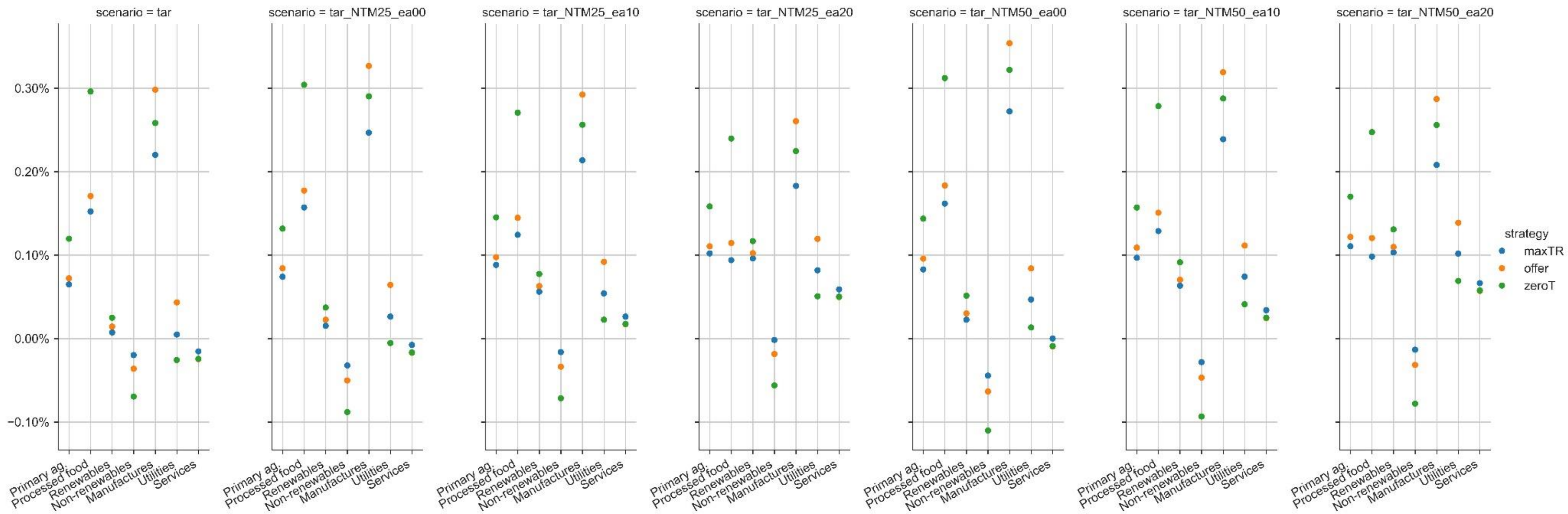


Source: the MAGNET calculations.

Figure 15 shows the effects on the output at the AU level in the main commodity categories. Most categories have at least some increases in the output. The non-renewables are the constant exception that experiences a small reduction in almost all cases. This indicates that the AfCFTA will affect Africa's traditional output mix. Further evidence is that the manufactures most consistently perform the best along with processed foods. The NTM cuts also boost the output of the manufactures the most. Thus, the AfCFTA helps Africa to both industrialize and become less dependent on non-renewable resource exports. Somewhat surprisingly, the services output decreases in cases when there is no or little spillover. While spillovers increase the output of renewable and non-renewable resources, utilities, and services, they also simultaneously dampen the increase of the processed foods and manufactures output. The increase in output of primary agricultural commodities is the steadiest across the cases.

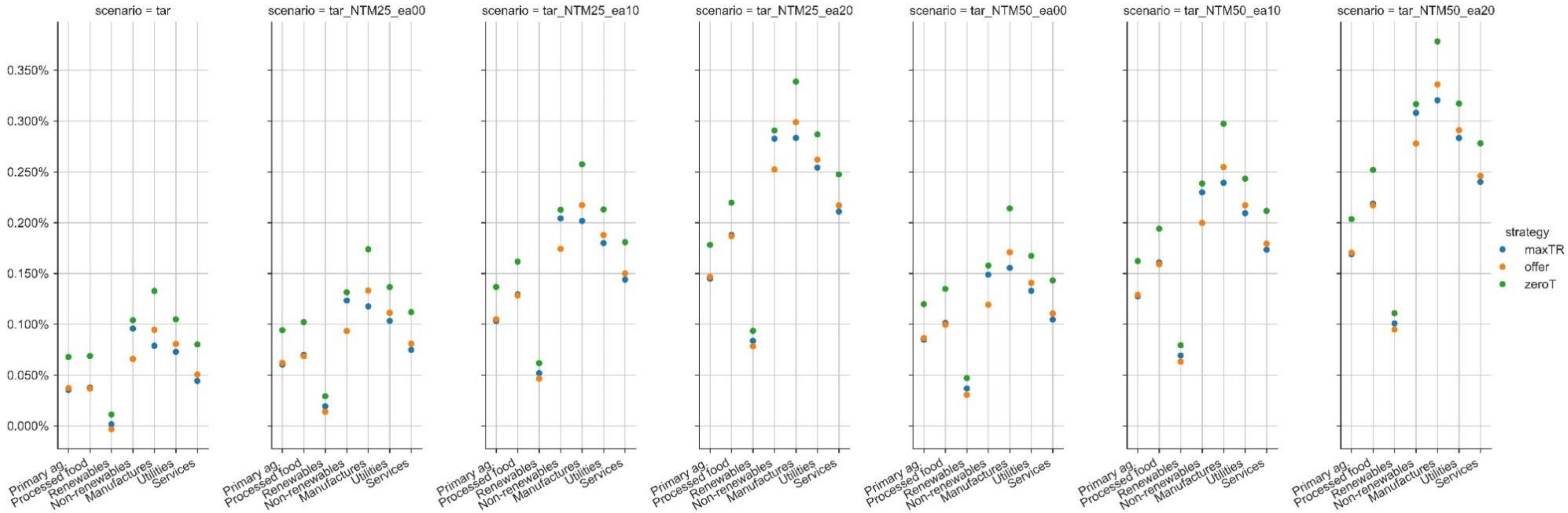
Figure 16 displays the changes in private consumption at the AU level in the main commodity categories. Consumption increases consistently with increasing incomes. Consumption also moves to higher value-added commodity groups along increasing incomes. For instance, the effect on primary agricultural commodities is similar when only tariff cuts are considered, but when we assume NTM reductions and spillovers, we find larger increases in processed foods consumption. The consumption of the manufactures increases the most in all cases. The results highlight again the disproportionate effect of the spillover assumption: the increases are relatively more significant for each change in spillover shocks when compared to the changes in the NTMs themselves.

Figure 15. Economic effects at the AU level - output by main commodity categories



Source: the MAGNET calculations.

Figure 16. Economic effects at the AU level - private consumption by main commodity categories



Source: the MAGNET calculations.

Regional level results

This section examines the regional variation in the results, presenting results as box plots of single region results depicting the policy simulation deviations from the baseline in 2035 when the AfCFTA is fully implemented. The mid-line in each box marks the median result while box edges mark the lower and upper quartiles of the distribution. The whiskers extend to 1.5 times the inter-quartile range (IQR). We omit the outliers to preserve the interpretability of all the figures.[9] The AU level results of which some were described above, are marked by 'x'.

We highlight one strategy-scenario combination (current tariff offers with NTM shocks and 10% NTM spillover) as the central scenario, and that scenario is also depicted as a heatmap for each African region in the model. The results are segmented into macro-level impacts, trade flow changes, and sector-specific effects.

Macro level impacts

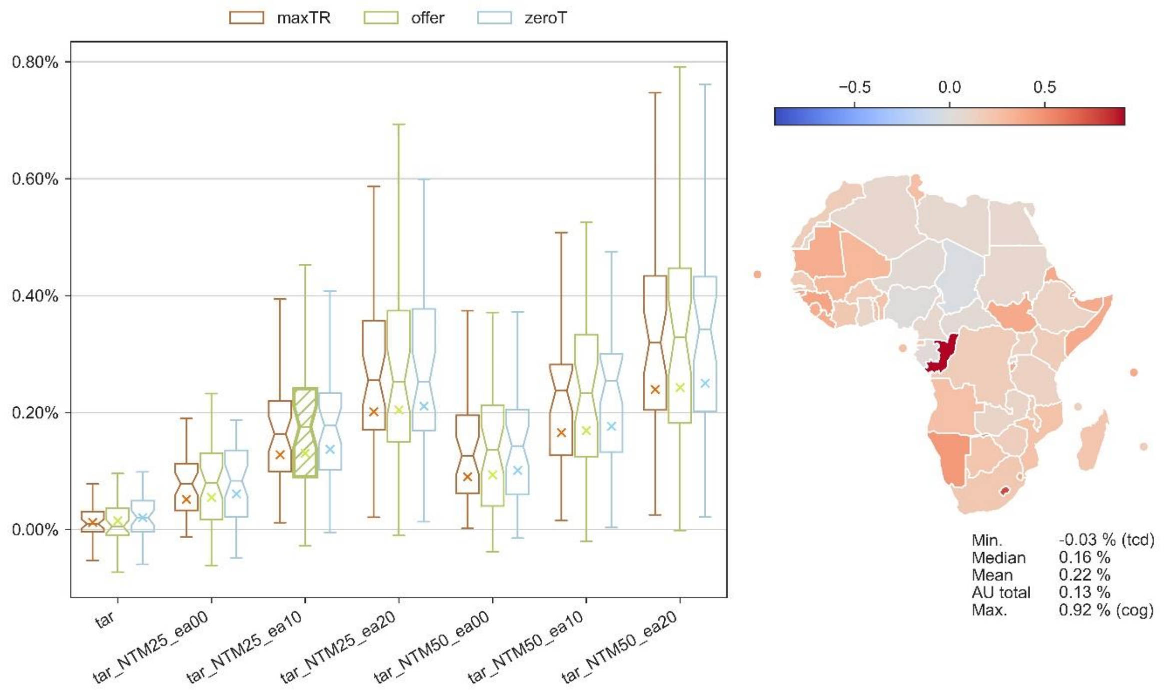
The implementation of the AfCFTA policies generally lead to an increase in trade volumes, particularly within African countries, which is an expected outcome of reducing trade barriers. The increased trade opportunities lead to higher incomes at the national and household levels.

Gross Domestic Product (GDP)

Figure 17 depicts the changes in GDP for each AfCFTA policy scenario. A detailed analysis of GDP changes indicates growth across most regions involved in the AfCFTA. The median GDP increase in our central scenario (current tariff offers with 25% NTM reduction and 10% NTM spillover) is 0.16%. The regions with the most significant GDP growth were Rest of SACU and Congo, with increases less than 1.0%. While tariff cuts alone have a marginal positive impact on GDP, the inclusion of NTM reductions contributes to a more substantial GDP growth, with additional spillover effects amplifying this further. The distribution of the outcomes also widens when the NTMs and spillovers are included, highlighting fewer universal capabilities in different regions of enjoying additional benefits brought by the reduction in NTMs.

[9] The complete results including the statistical outliers can be accessed in the dashboard.

Figure 17. The AfCFTA effects on GDP (percentage change from baseline in 2035)

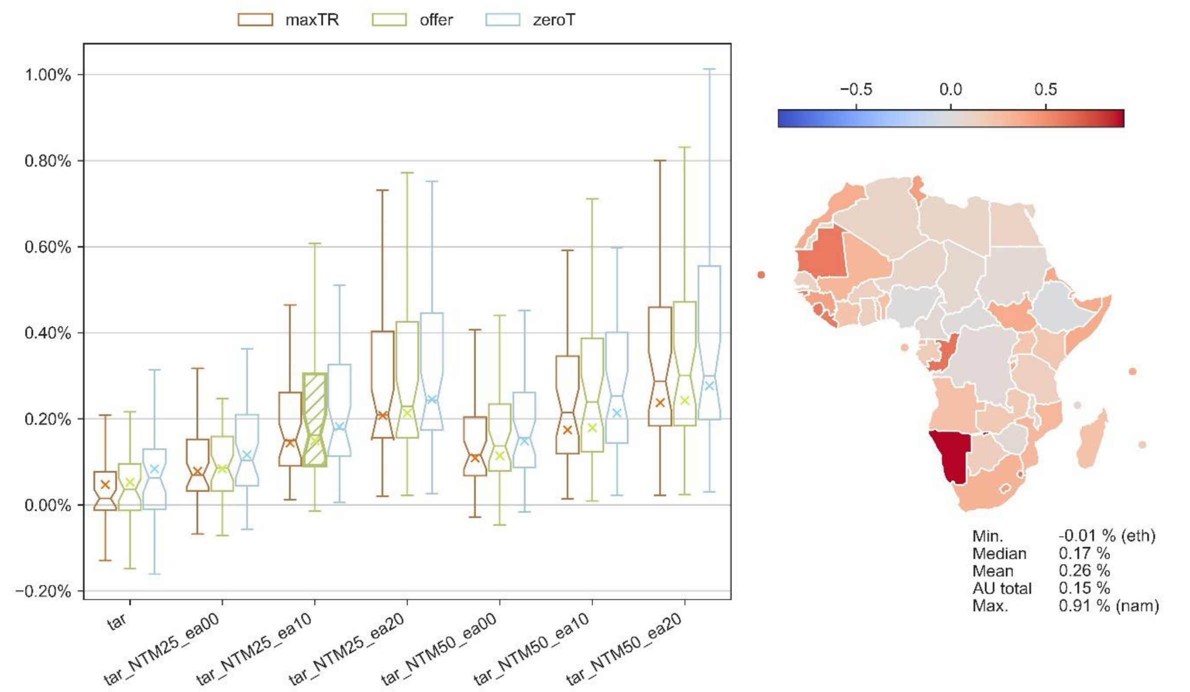


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Household disposable income

The effects on household disposable income largely mirror the GDP ones, with most regions experiencing positive changes (Figure 18). The median effect is slightly higher, 0.26%, which means that households gain a higher income increase than the other sectors. The variation in disposable income across different policy scenarios emphasizes the importance of the NTM reductions, which appear to be a significant driver of income changes. Namibia has the highest increase, 0.91%.

Figure 18. The AfCFTA effects on household disposable income (percentage change from baseline in 2035)

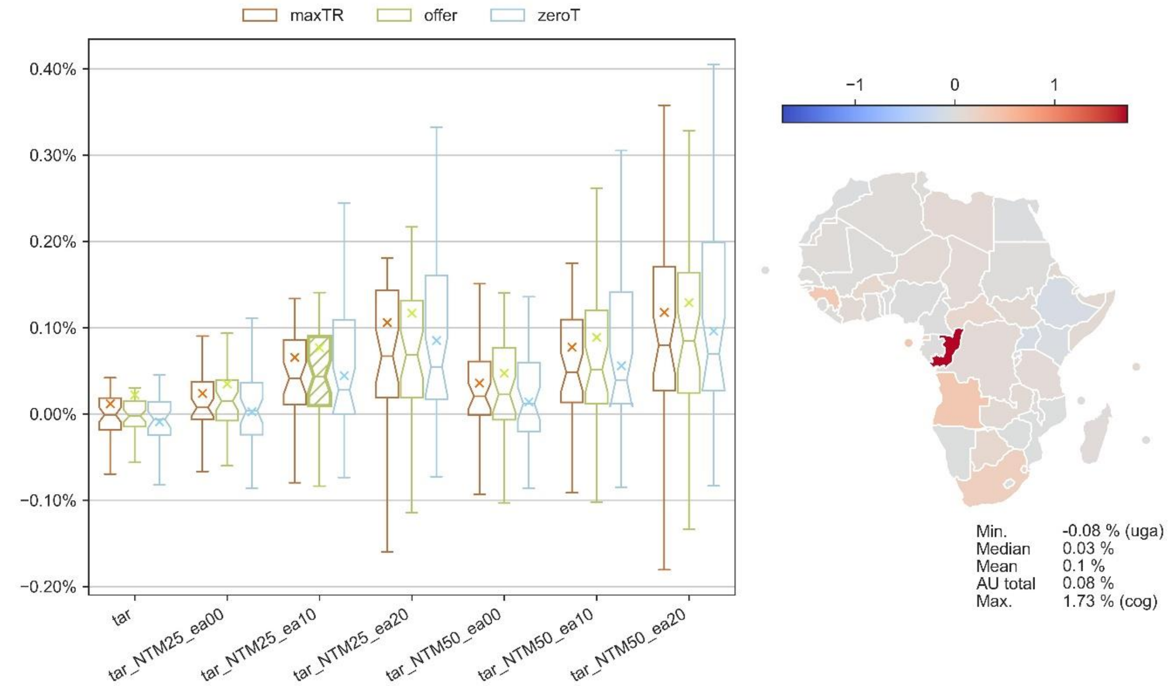


Source: the MAGNET calculations (n note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Investments

Figure 19 shows the effects on real investment demand. The AU level increase is lower than other absorption items, and the increase in investments is much more regionally dispersed. In our preferred scenario the AU total (0.08%) is higher than the median increase (0.03%). There are several regions where investments decrease with respect to the baseline, but the increase in South Africa (0.28%), the largest African economy, is enough to increase the AU total and mean effects. The highest increase happens in Congo (1.73%), while investments decrease the most in Uganda, 0.08%.

Figure 19. The AfCFTA effects on real investment demand (percentage change from baseline in 2035)

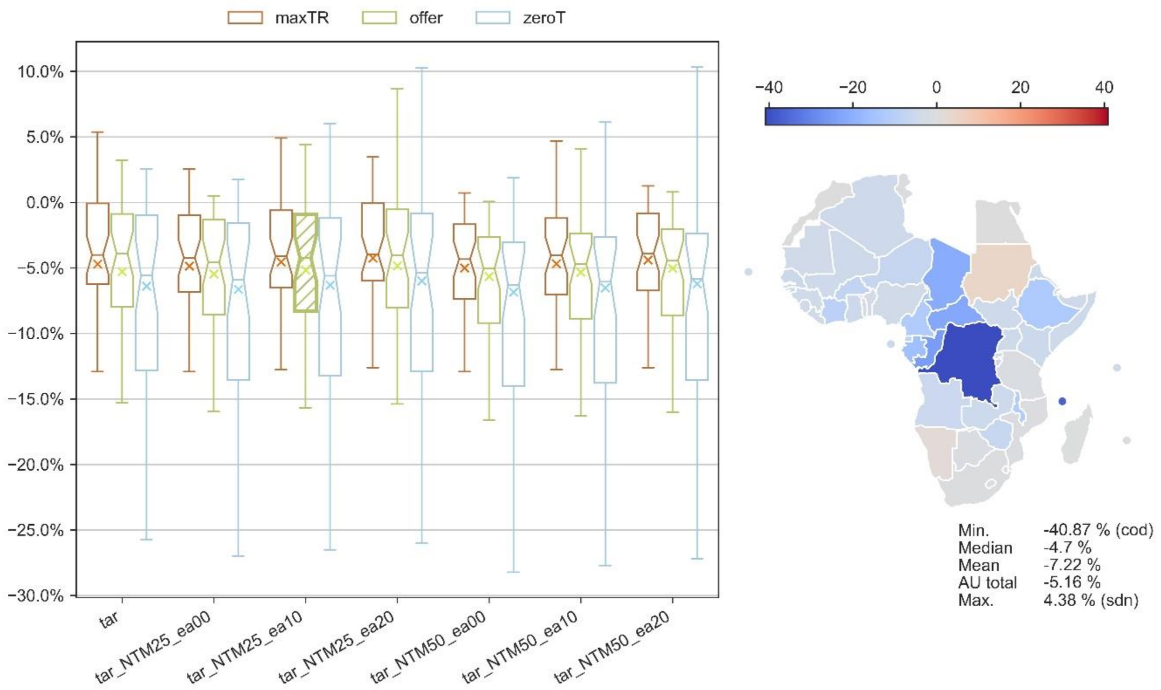


Source: the MAGNET calculations (n note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Public spending and tariff revenue

A reduction in tariff incomes is a direct consequence of trade liberalization (Figure 20). The estimated decrease in tariff revenue is most pronounced in the scenario of full trade liberalization, where countries retain only the revenue from tariffs imposed on non-African countries. In our preferred scenario the largest reduction occurs in the Democratic Republic of Congo (-41%) while the median reduction is -4.7%.

Figure 20. The AfCFTA effects on tariff revenue (percentage change from baseline in 2035)[10]

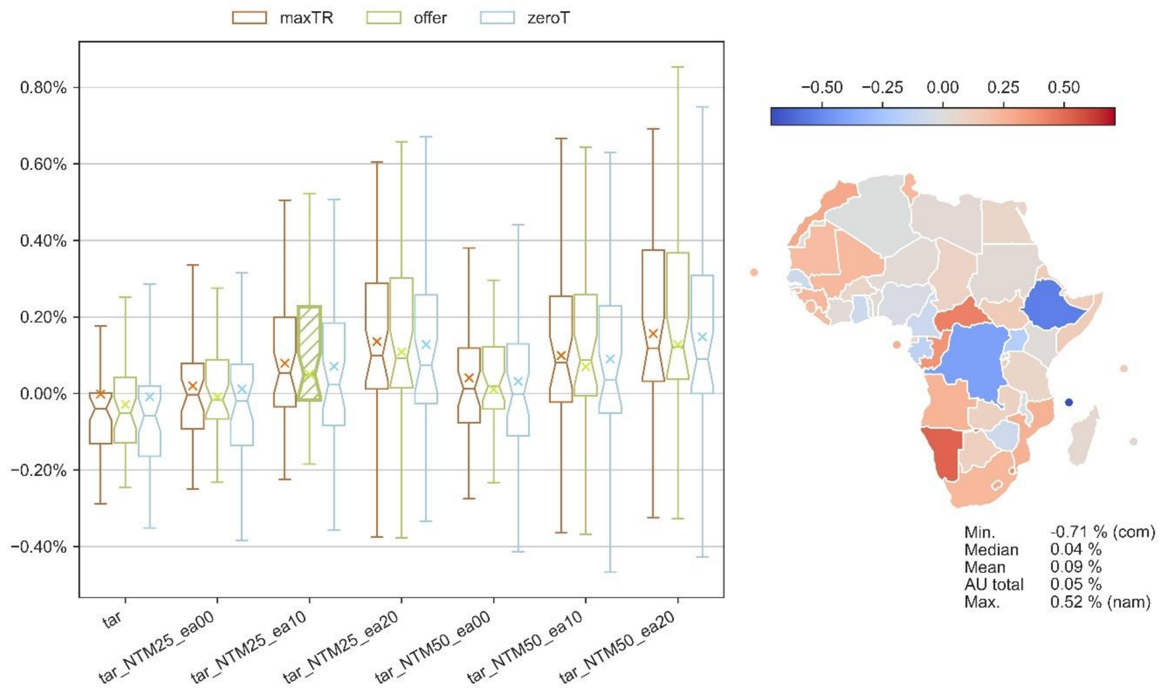


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

However, domestic tax income is likely to increase due to increased economic activity, potentially offsetting the loss in tariff revenue. Figure 21 depicts the overall change in public spending due to the AfCFTA. The results are mixed on regional level, although the median effect is slightly positive (0.04%). The mere tariff cuts have a negative effect on public spending on median level, and the NTM effects are needed to maintain the public sectors from shrinking. The moderate 25% NTM reduction without spillovers doesn't yield positive change in public spending.

[10] The region Rest of North Africa doesn't have any tariffs in place in the database and therefore doesn't have tariff revenue changes due to the AfCFTA. The region includes Libya and West Sahara, both unstable regions at the database base year 2017.

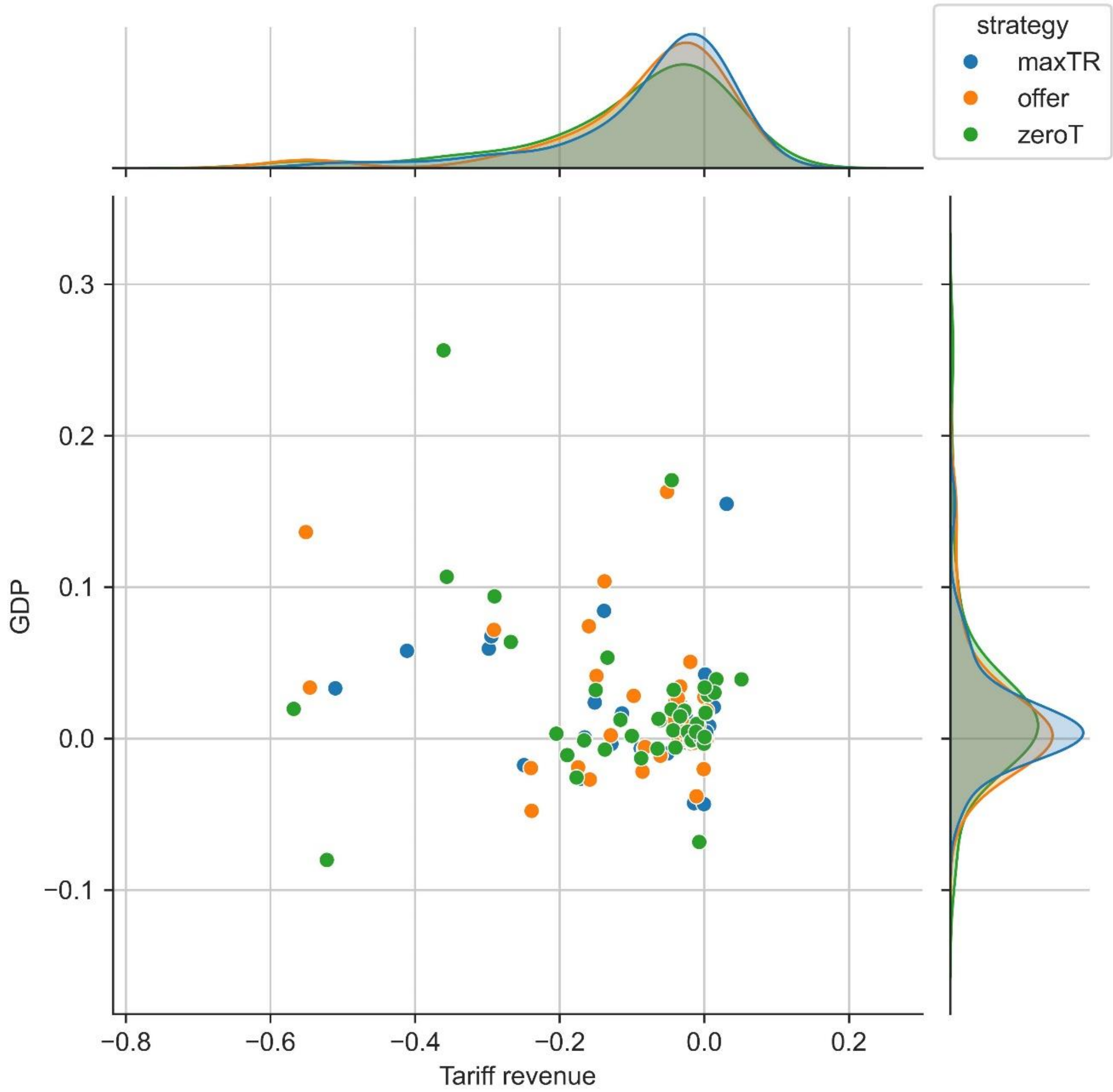
Figure 21. The AfCFTA effects on public spending (percentage change from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

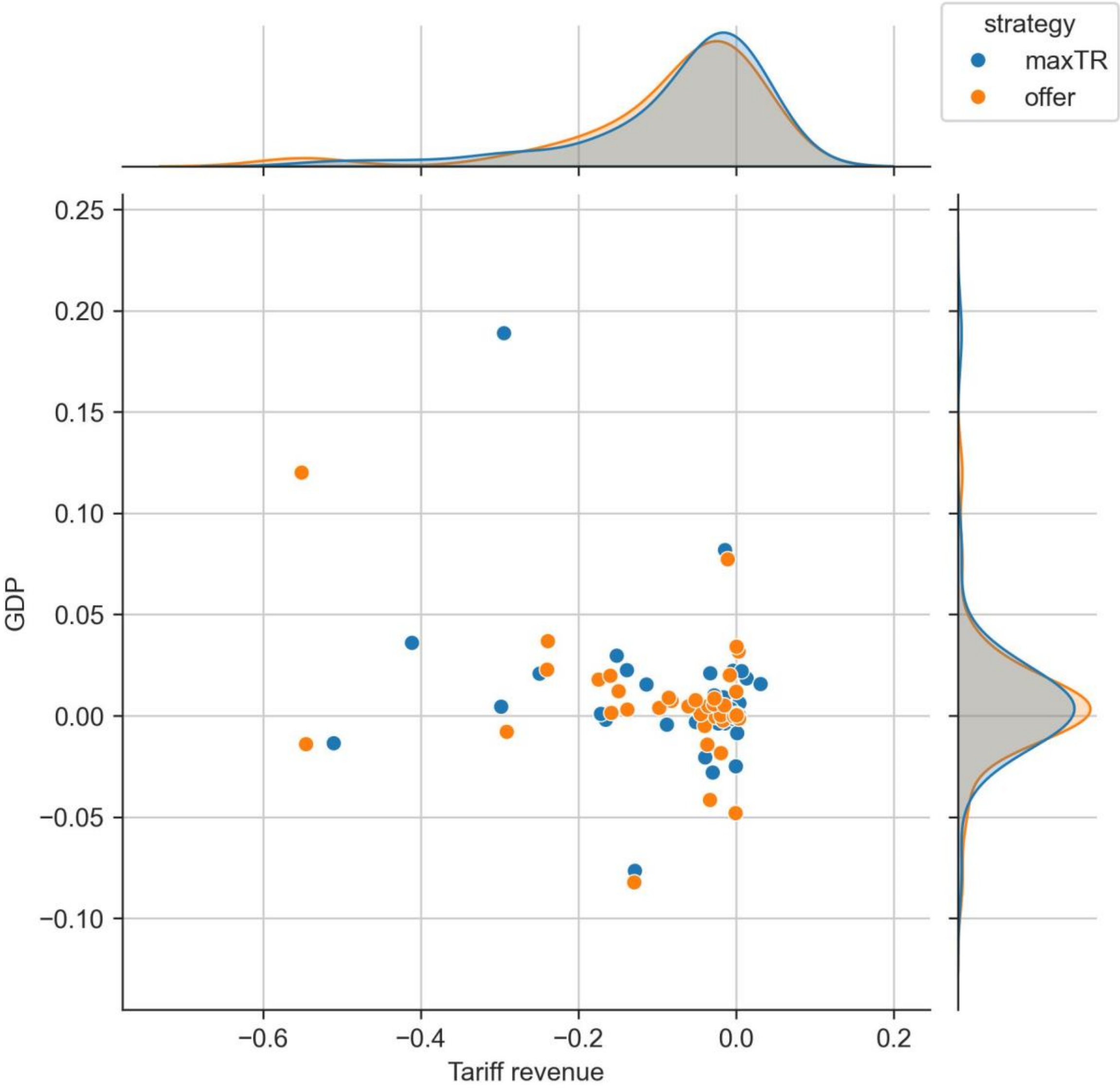
The tariff cuts only do not show unanimous evidence for economic benefits (Figure 22). In absolute changes, the GDP change is positive in majority of cases, but the losses in tariff revenues are larger in many cases. Thus, to balance the public sector budgets, additional domestic taxation is needed in these cases. Figure 23 illustrates how much the full liberalization would change the situation. We see that the effect is even stronger: small gains in GDP can be achieved by larger need for public sector balancing. The benefits from the NTM reductions are therefore essential for maintaining public sector in balance.

Figure 22. The AfCFTA effects on tariff revenue and GDP (absolute change from baseline in 2035 in billion USD)



Source: the MAGNET calculations.

Figure 23. The AfCFTA effects on tariff revenue and GDP - difference to full liberalization scenario (absolute change from baseline in 2035 in billion USD)



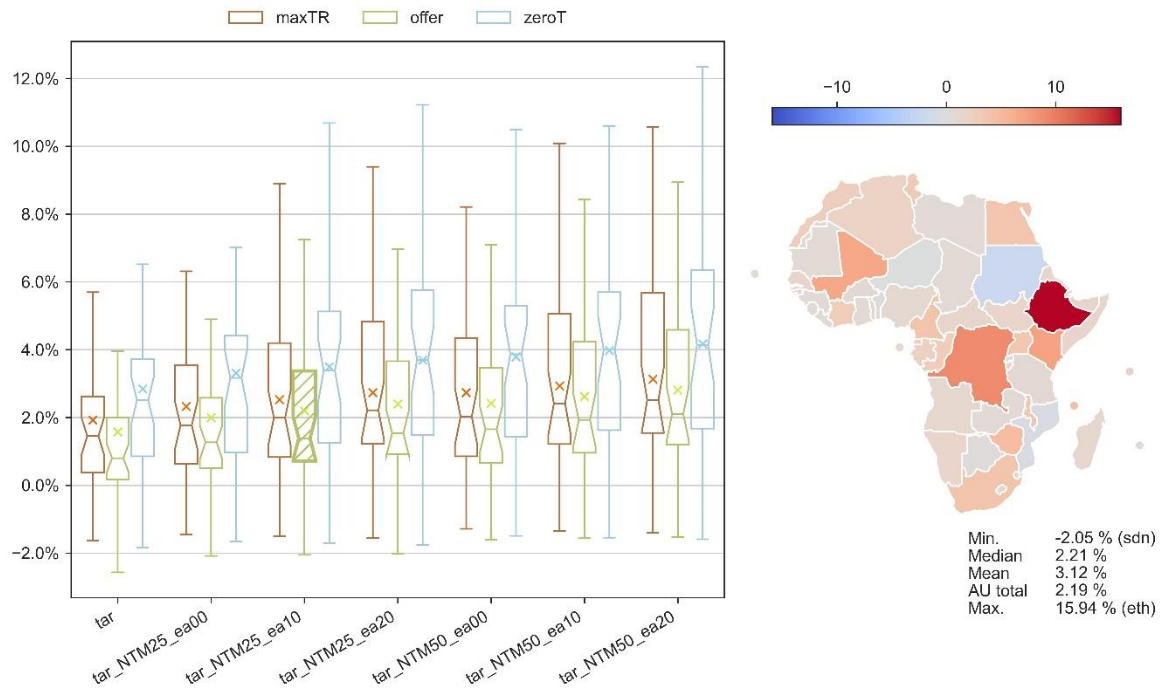
Source: the MAGNET calculations.

Trade flow changes

Overall trade

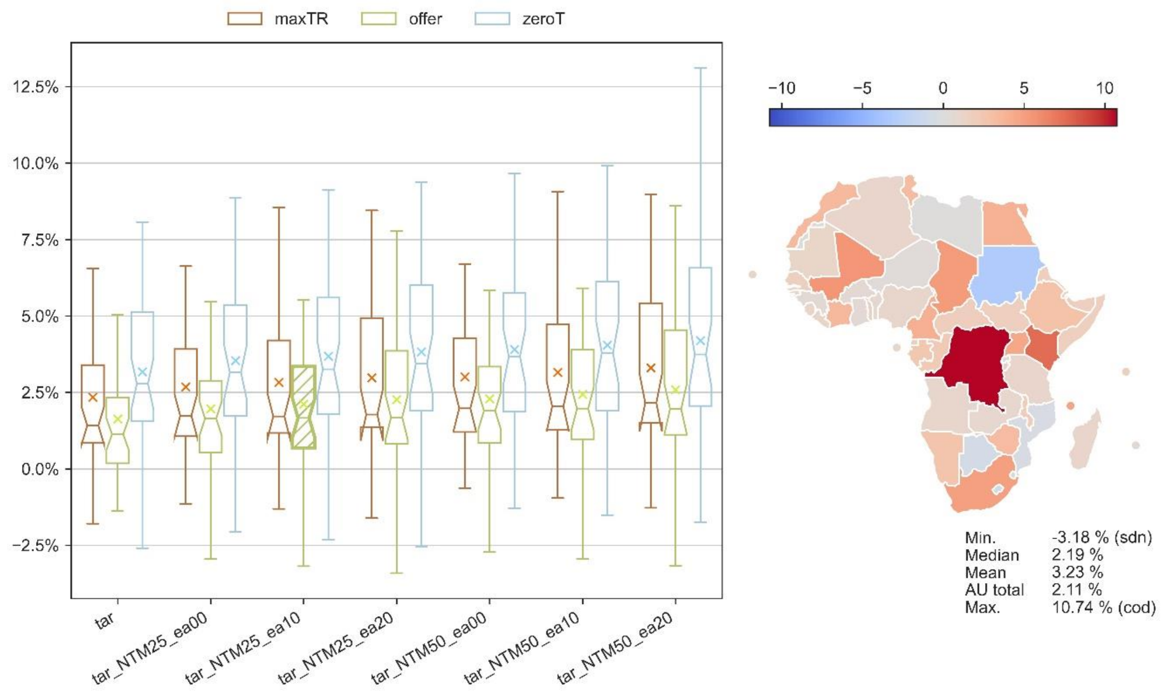
The AfCFTA leads to modest increases in total trade flows, aligning with the growth in GDP and consumption (Figure 24 and Figure 25 for exports and imports, respectively). Median increases in exports and imports are both approximately 2.2% in our preferred scenario. NTM reductions increase the median level effect, and the effect is also higher when there are extra-African spillovers involved.

Figure 24. The AfCFTA effects on total export volumes (percentage change from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 25 The AfCFTA effects on total import volumes (percentage change from baseline in 2035)

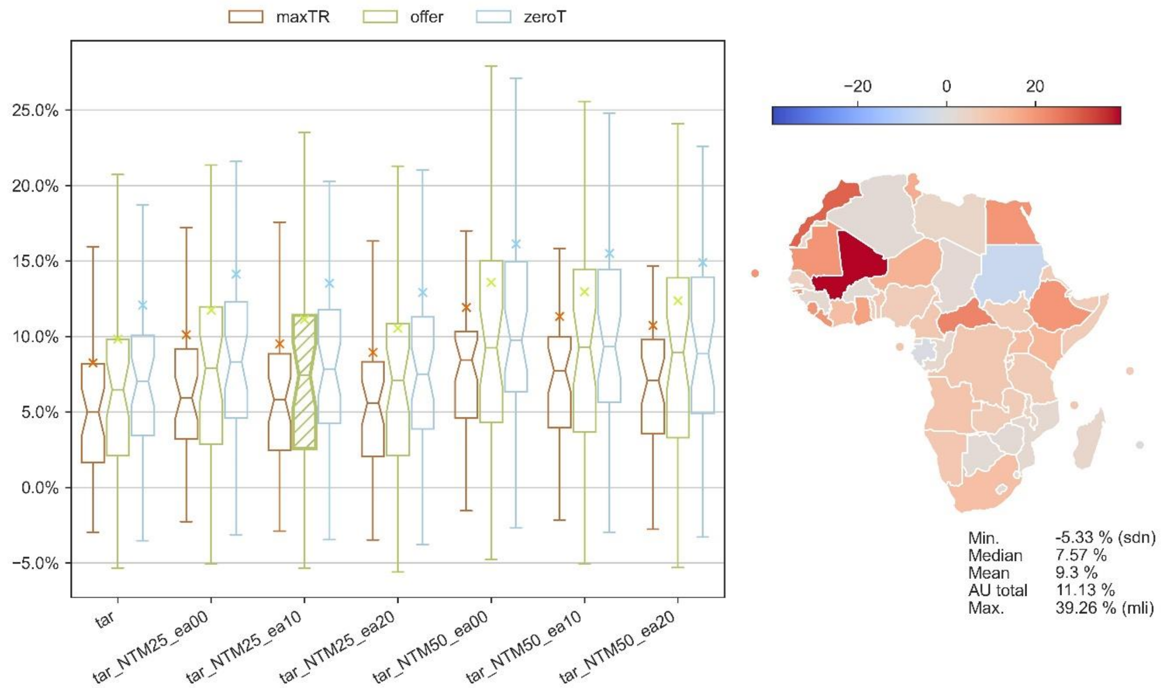


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Intra-African trade

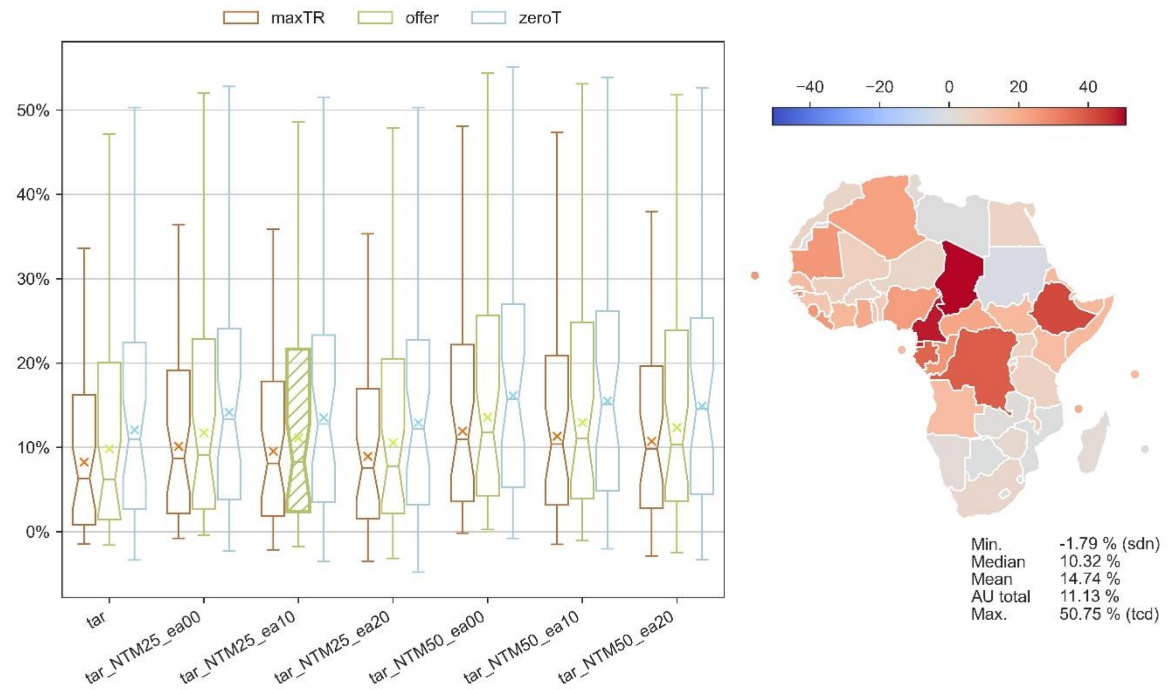
Notably, intra-African trade experiences a more pronounced increase, with median exports and imports growing by 7.6% and 10%, respectively (Figure 26 and Figure 27). This is indicative of the AfCFTA potential in bolstering regional economic integration. The shares of intra-African trade in the total trade of African countries also increase, which is one of the goals of the agreement. Figure 28 and Figure 29 illustrate the change for exports and imports, respectively. The median change is 1.0% point for both exports and imports. Whereas the extra-African spillovers generate higher overall trade-flows that is at the expense of intra-African trade, and we see small reduction of intra-African trade for higher spillover effects.

Figure 26. The AfCFTA effects on intra-African export volumes (percentage change from baseline in 2035)



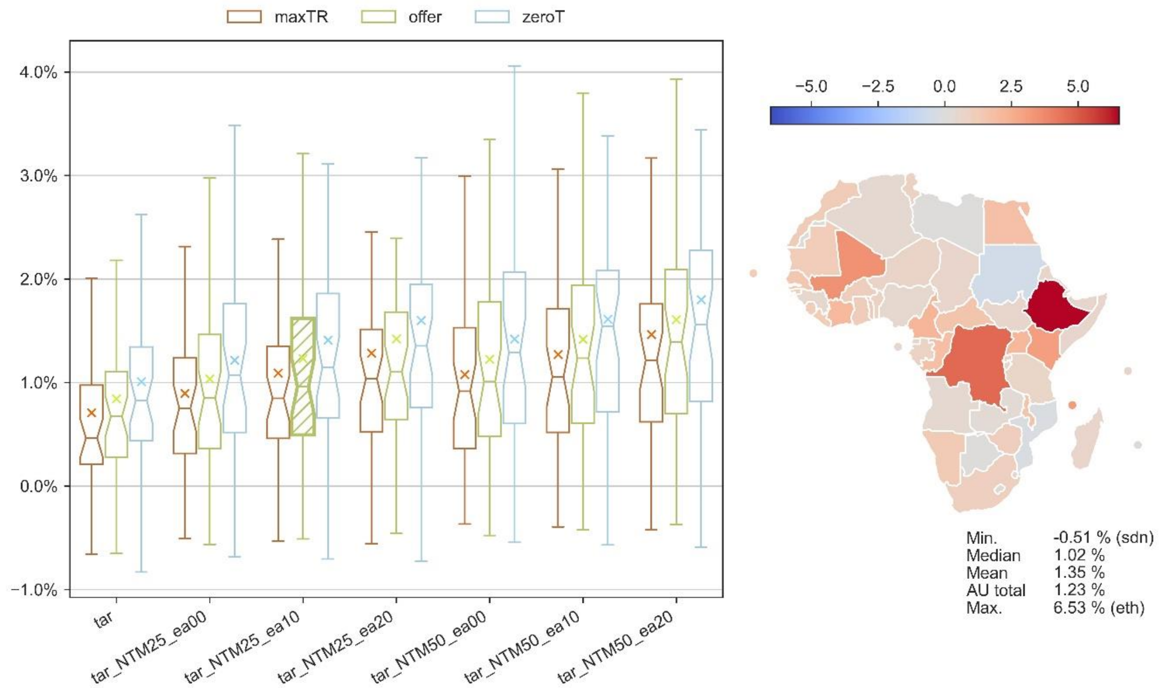
Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 27. The AfCFTA effects on intra-African import volumes (percentage change from baseline in 2035)



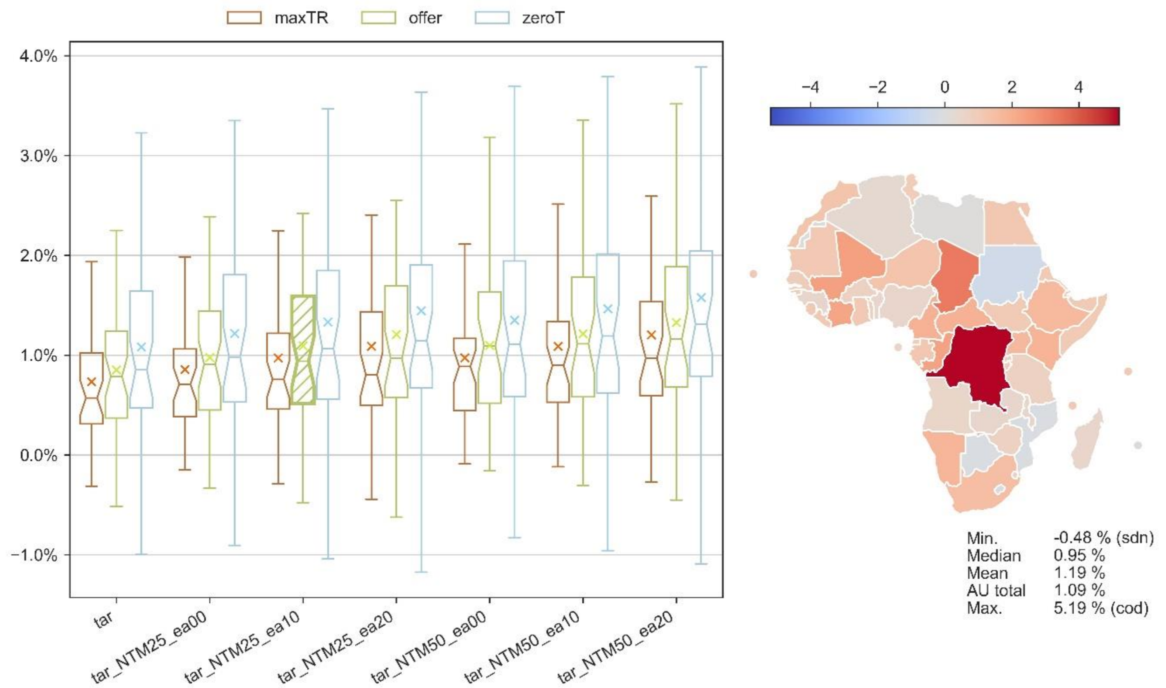
Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 28. The AfCFTA effects on intra-African export shares (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 29. The AfCFTA effects on intra-African import shares (percentage point deviation from baseline in 2035)

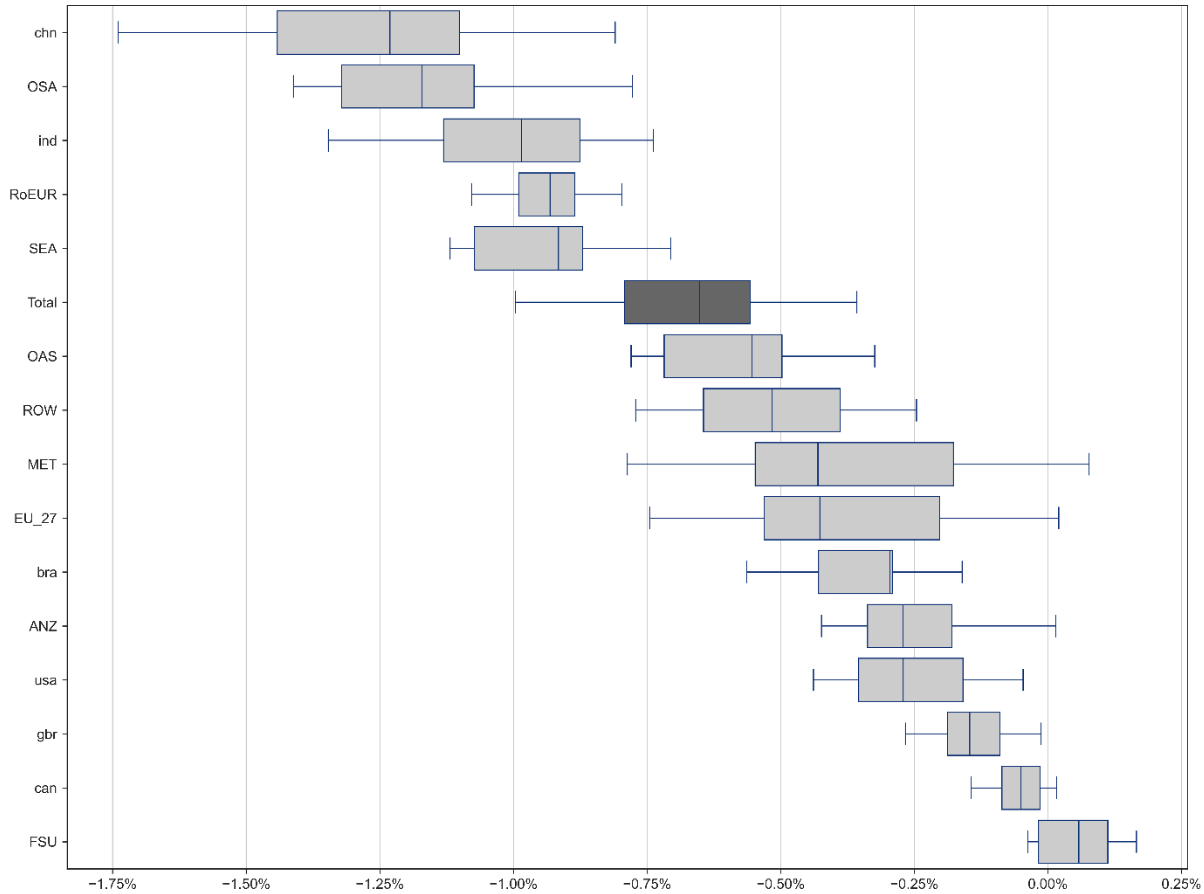


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Extra-African trade flows

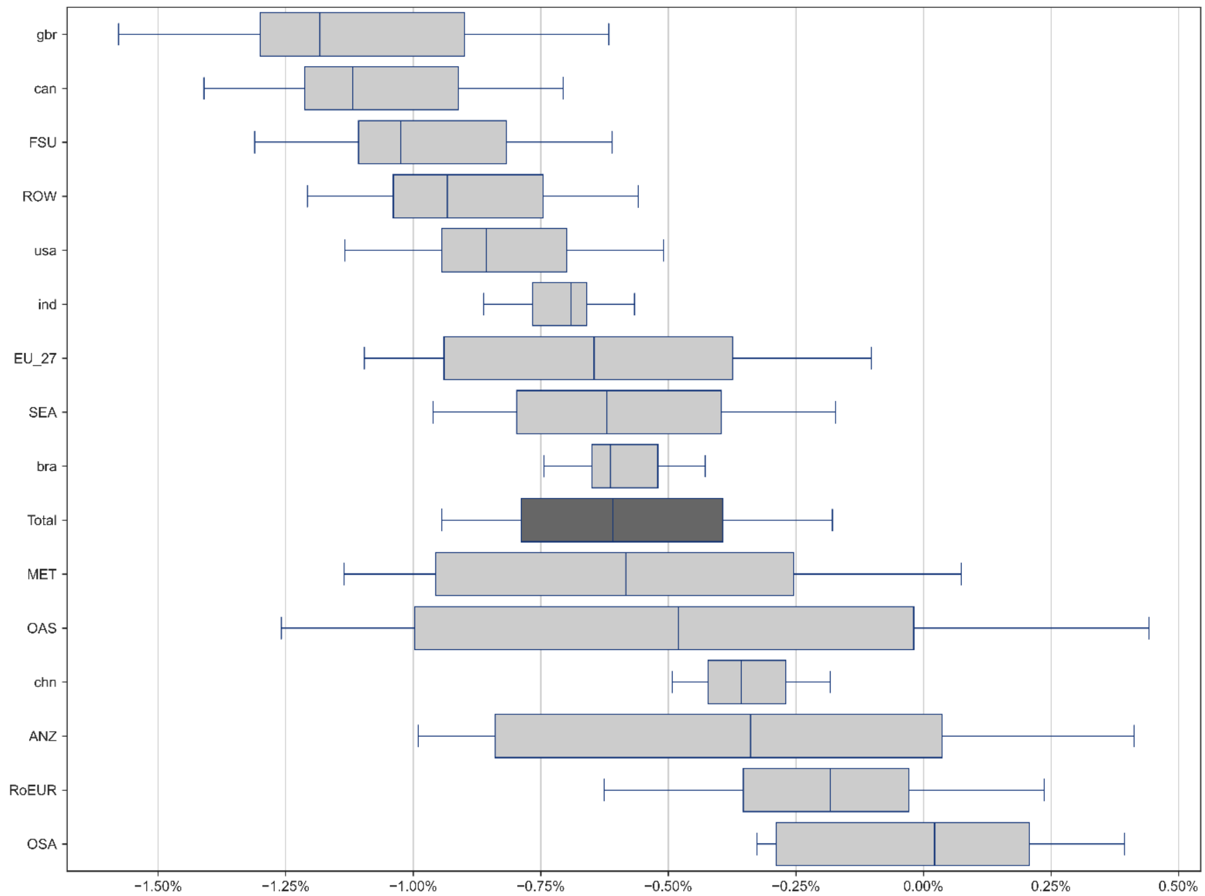
In each simulation intra-African trade flows will increase, and in almost every case the extra-African trade flows will decrease. Figure 30 and Figure 31 illustrate the percentage changes with respect to the baseline for extra-African exports and imports, respectively. Most non-African regions decrease their exports to Africa due to the AfCFTA. The only exception is the Former Soviet Union (FSU) that has a small increase. The highest decreases appear for China (chn), Other South America (OSA) and India (ind). In imports the regions with highest decreases are the Great Britain (gbr), Canada (can) and the Former Soviet Union (FSU).

Figure 30. Changes in extra-African exports by destination (percentage change from baseline in 2035)



Source: the MAGNET calculations (note: the box plot shows the variation among all the policy scenarios).

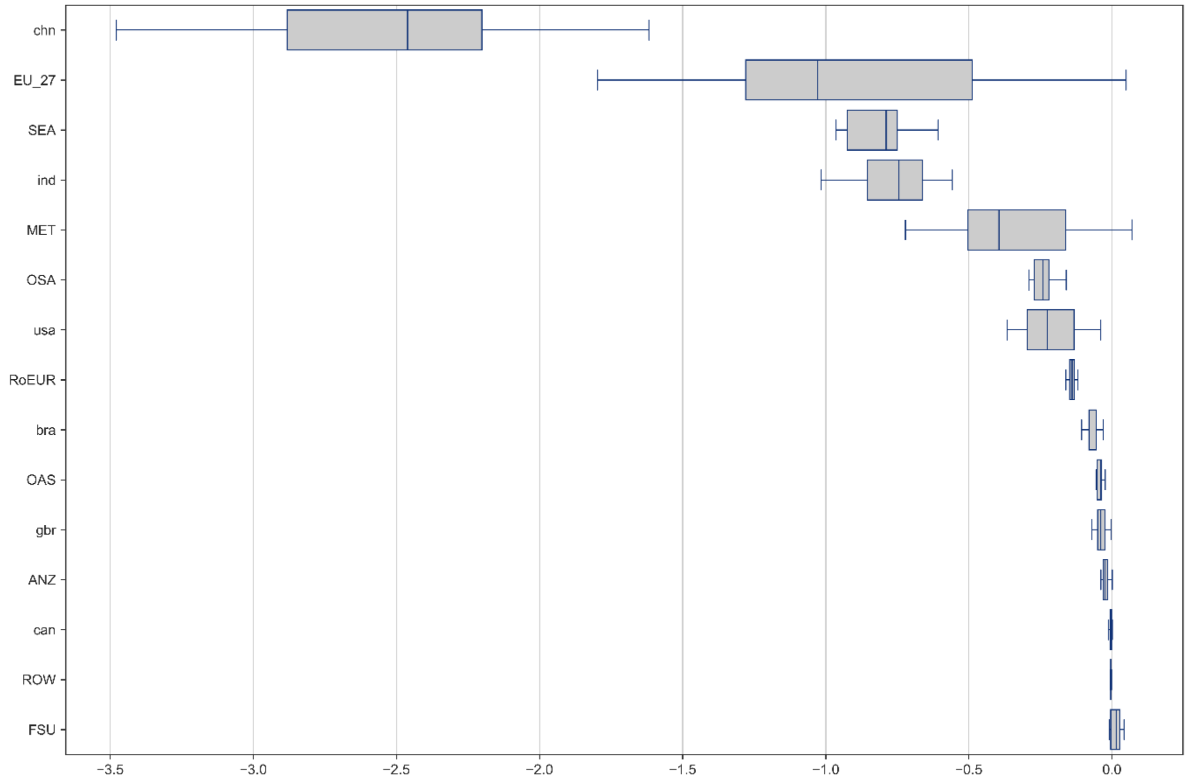
Figure 31. Changes in extra-African imports by source (percentage change from baseline in 2035)



Source: the MAGNET calculations (note: the box plot shows the variation among all the policy scenarios).

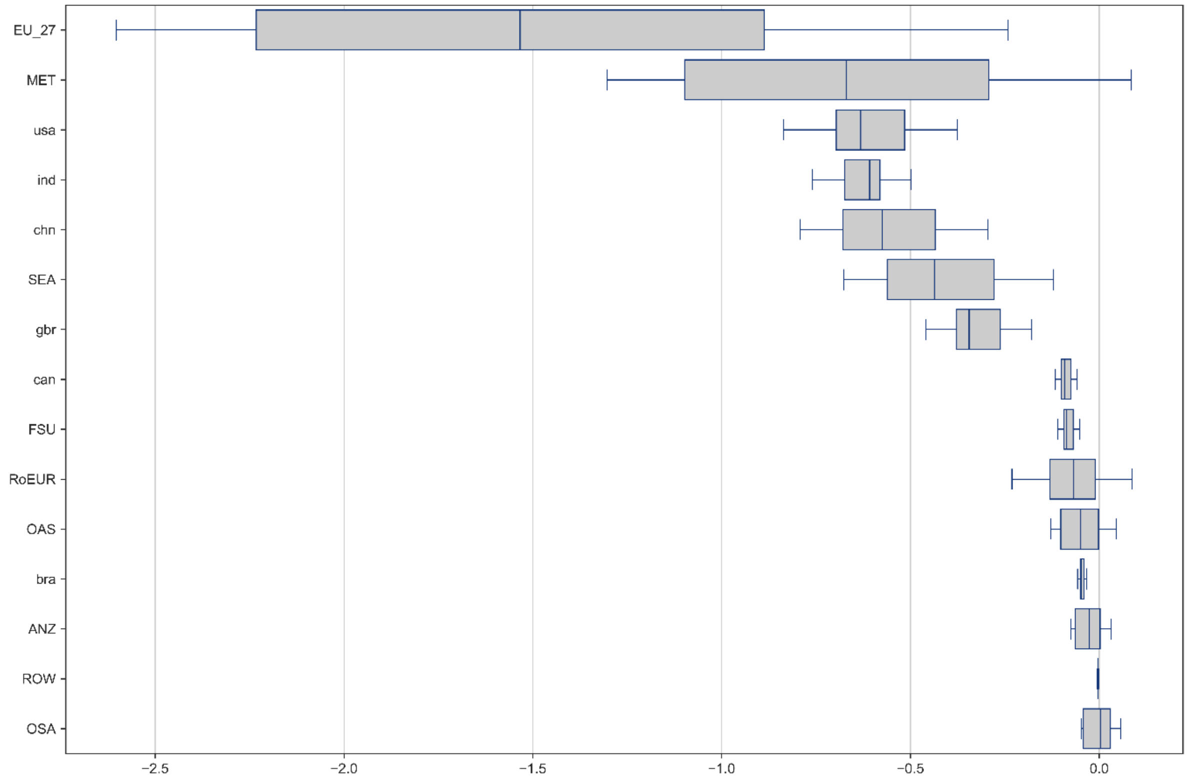
Figure 32 and Figure 33 show the same changes, but in absolute terms. Now China and the EU (EU_27) have the highest decreases in exports, while the EU and the Middle-East (MET) have the highest decreases in imports. The relative increase in FSU, appears rather marginal in absolute terms.

Figure 32. Changes in extra-African exports by destination (absolute change from baseline in 2035; billions of 2017 USD)



Source: the MAGNET calculations (note: the box plot shows the variation among all the policy scenarios).

Figure 33. Changes in extra-African imports by source (absolute change from baseline in 2035; billions of 2017 USD)



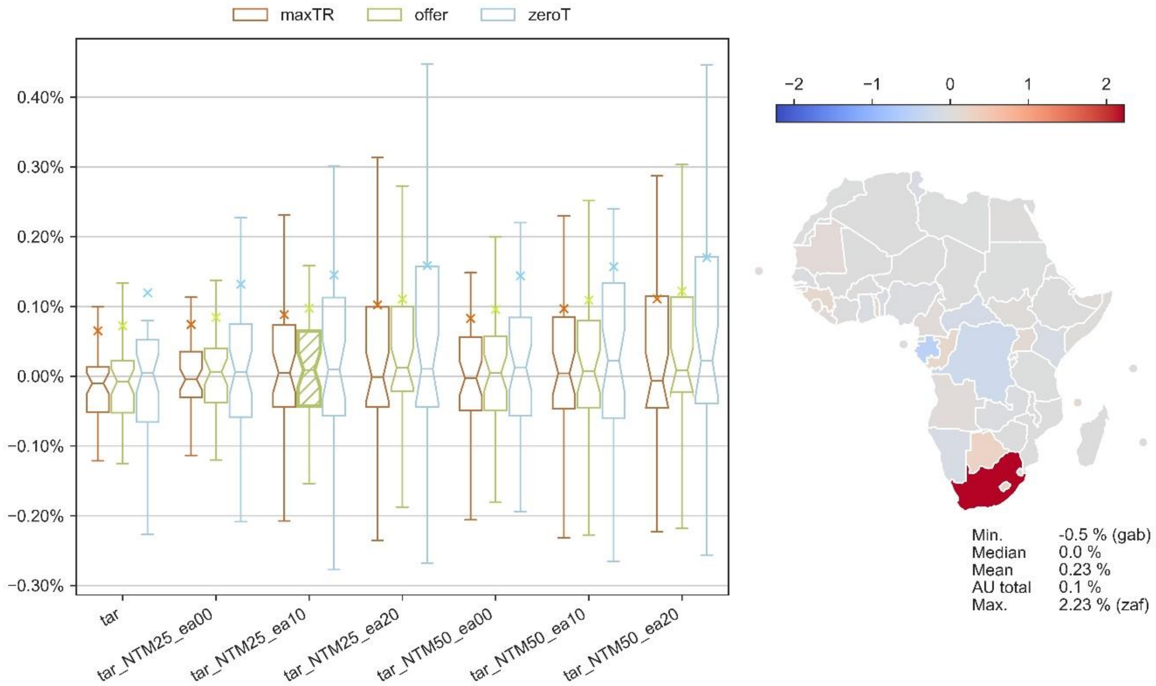
Source: the MAGNET calculations (note: the box plot shows the variation among all the policy scenarios).

Sector-specific effects

An in-depth analysis of sector-specific outcomes is necessary for understanding the nuanced effects of the AfCFTA on different segments of the economy. For instance, the agri-food sector, industrial manufacturing, and services sectors may respond differently to the policy changes. By examining these sectors, we can better comprehend the implications for employment, value chain development, and regional specialization.

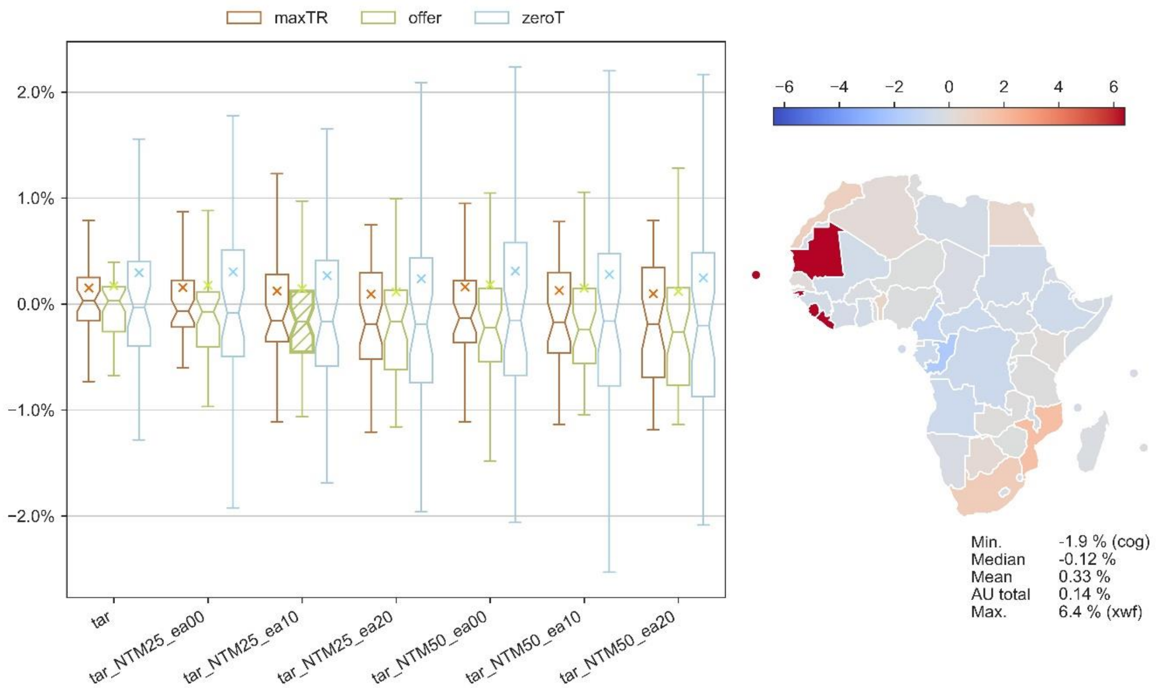
Primary agricultural production increases in several regions while the mean increase is a moderate 0.2% and median change is zero (Figure 34). Regional disparities are large, and South Africa is set to increase its primary agricultural sector the most, by 2.2%. The processed food output increases somewhat more, 0.3% on average although with considerably more regional variation (Figure 35). Despite an average increase, there is a median decrease in processed foods output, which means that more regions decrease their output. Thus, the AfCFTA increases the concentration of food processing in the continent. The increase is highest in the Rest of Western Africa region, 6.4%.

Figure 34. The AfCFTA effects on primary agricultural outputs (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

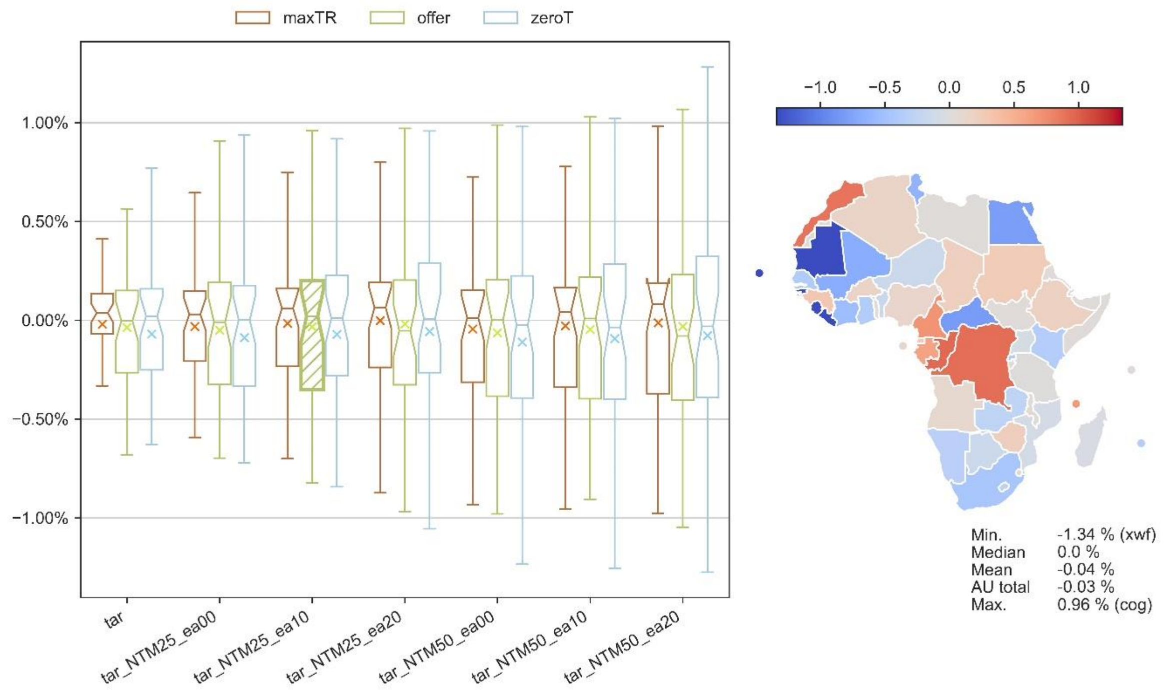
Figure 35. The AfCFTA effects on processed food outputs (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

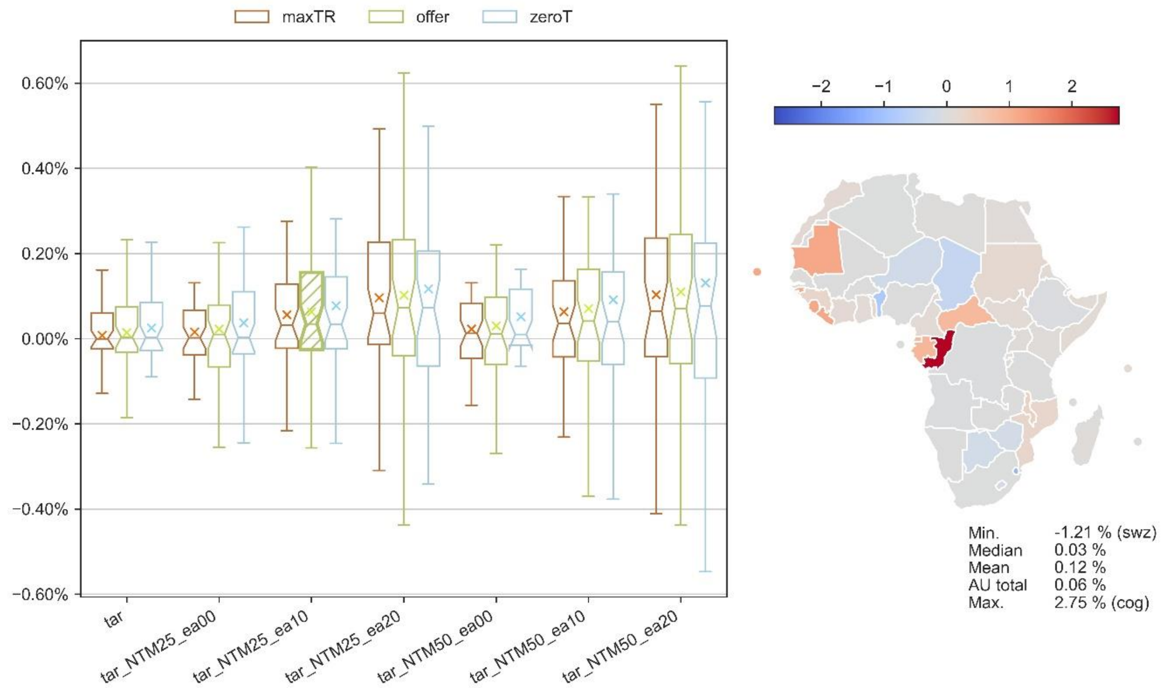
Figure 36 and Figure 37 show the changes for non-renewable and renewable resource outputs, respectively. Despite a heavy reliance on natural resource exports, the non-renewable resource output is very little affected by the AfCFTA because of the exports being mainly directed to non-African countries. There is however considerable regional variation: in our preferred scenarios Congo, Democratic Republic of Congo and Morocco increase their output between 0.9% and 1.0%. The renewable resource output increases slightly more than the non-renewable resource output due to more demand being within Africa.

Figure 36. The AfCFTA effects on non-renewable resource outputs (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 37. The AfCFTA effects on renewable resource outputs (percentage point deviation from baseline in 2035)

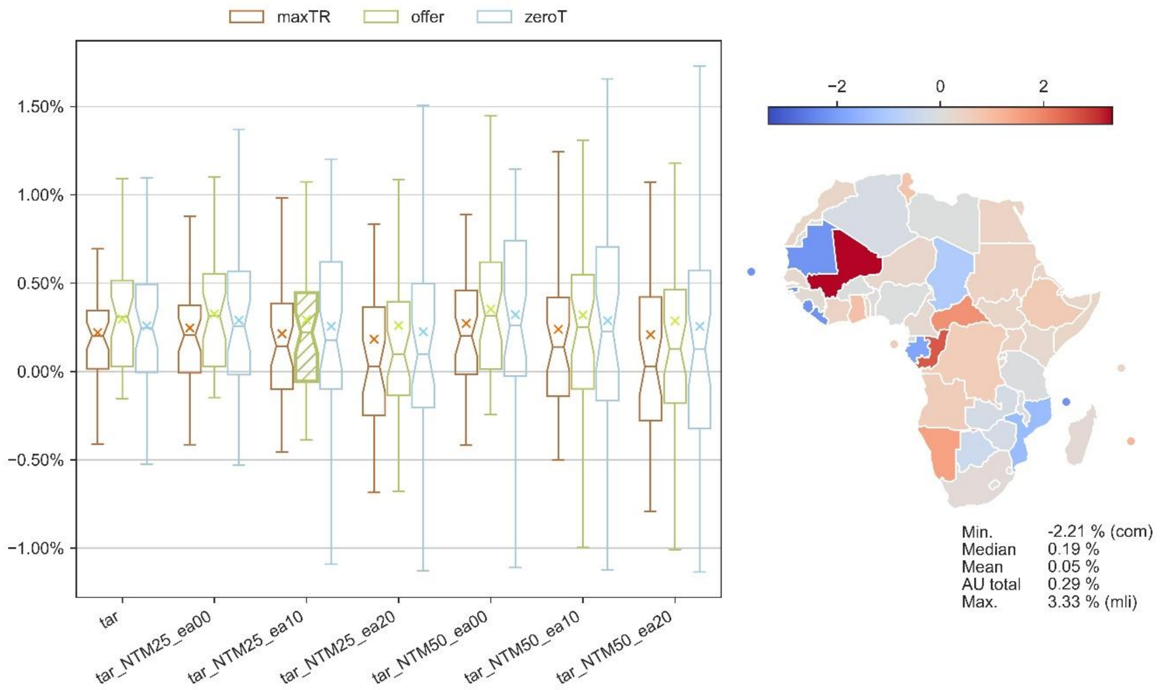


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 38 shows the effects on the manufacturing sector outputs. We see that of all the sectors, it is the one that increases the most in the AU level, 0.29% in our preferred scenario. This is a positive result with respect to the AU goal of advancing industrialization in Africa. However, regional variation is relatively high. The spillover effect slightly dampens the outputs of the manufactures.

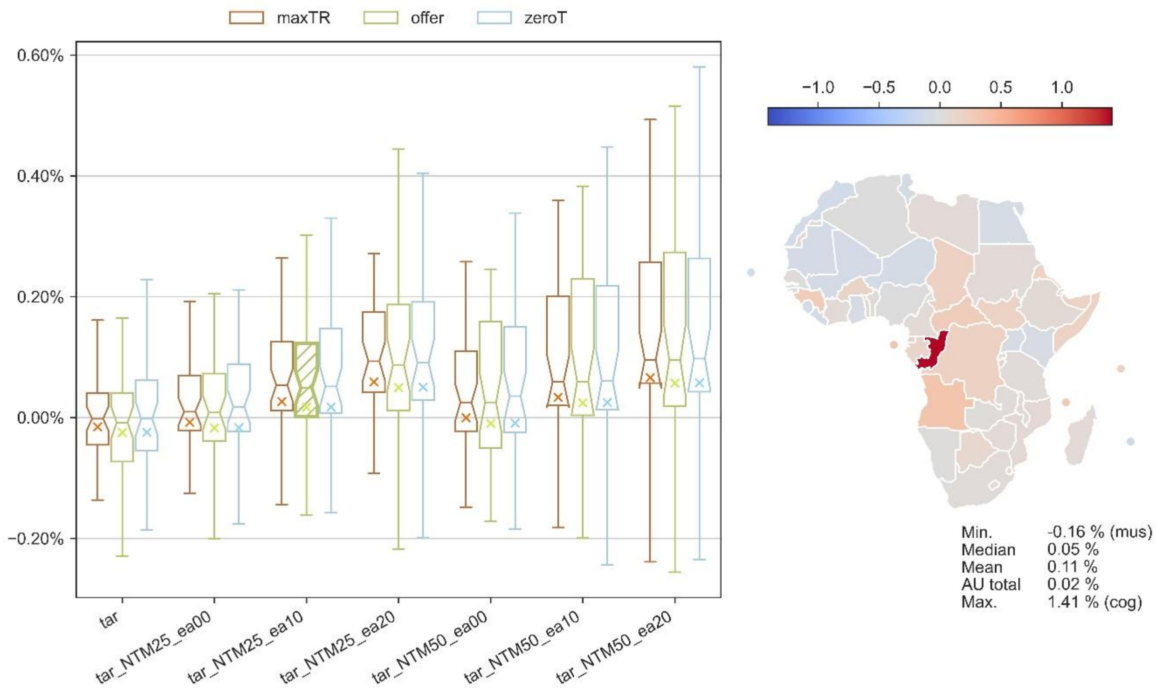
Figure 39 shows the effects on the services sector outputs. The mean change is only slightly positive at the AU level. However, there are notable regional differences for services as well. In sum, the AfCFTA is promoting more industrialization than a move to service economy. In addition, we can see that the NTM spillovers benefit the services at the same time when they dampen the increase in manufacturing.

Figure 38. The AfCFTA effects on manufacturing outputs (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 39. The AfCFTA effects on services outputs (percentage point deviation from baseline in 2035)

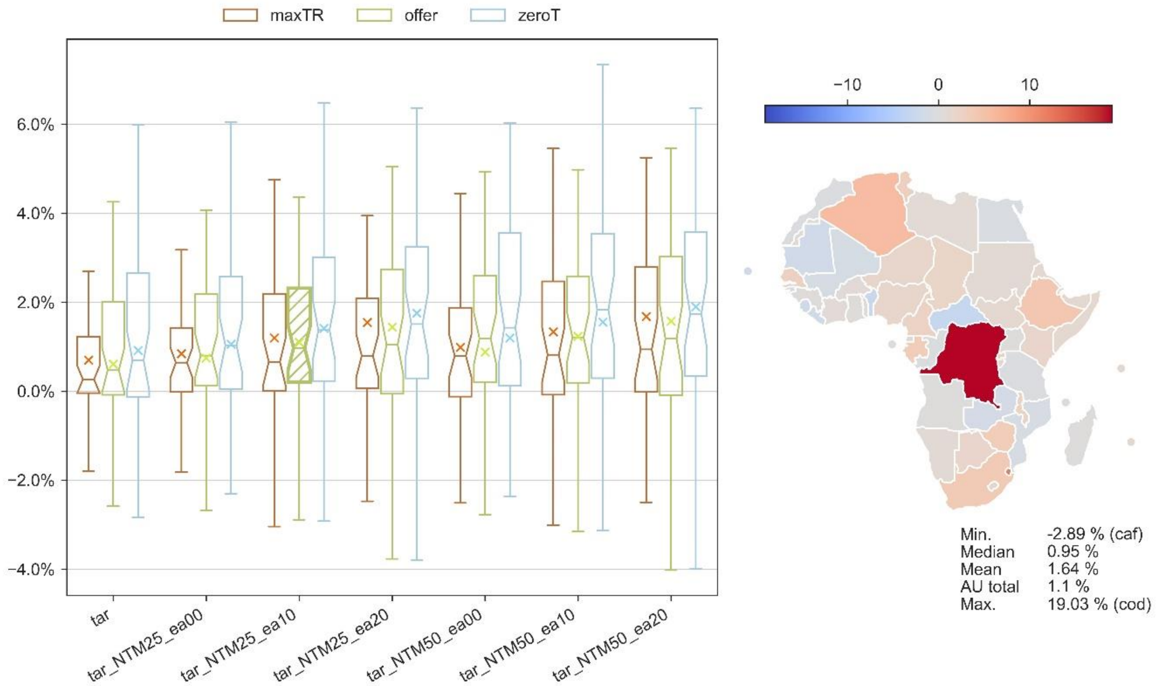


Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Effects on trade flows by commodity

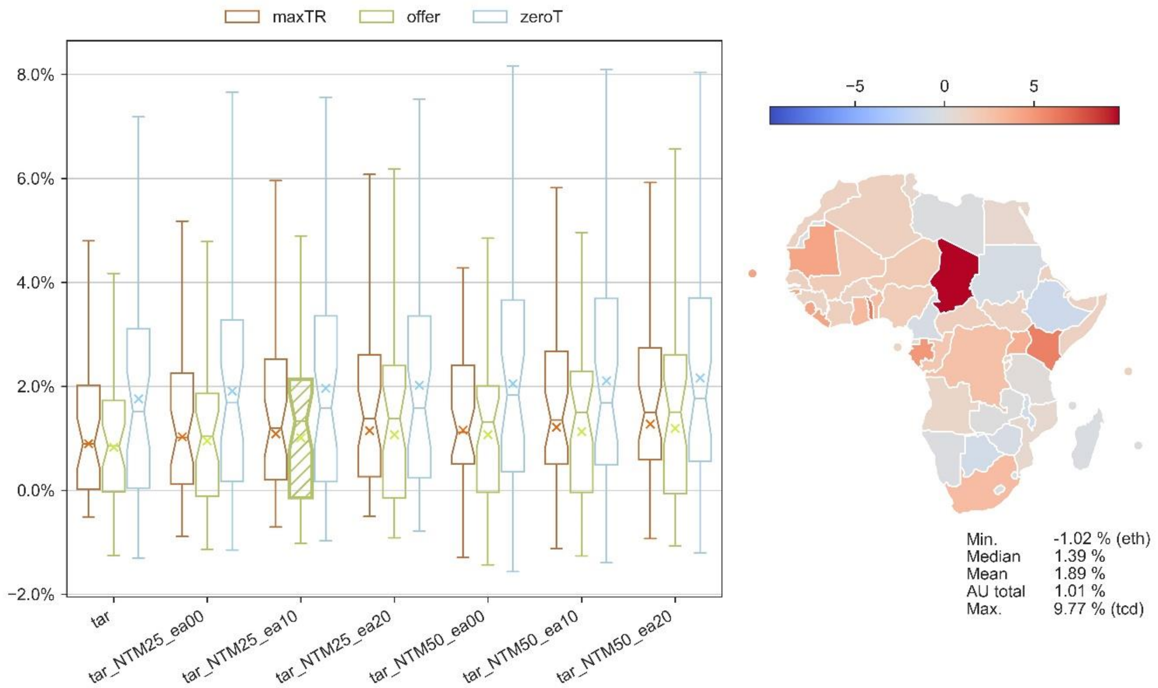
This section summarizes the changes in trade flows by commodity groups. Both exports (Figure 40) and imports (Figure 41) of the primary agricultural commodities increase on average, but with some regional variation. The highest increase in exports occurs in Democratic Republic of Congo (19%) while the imports increase the most in Chad (9.8%).

Figure 40. The AfCFTA effects on primary agricultural exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

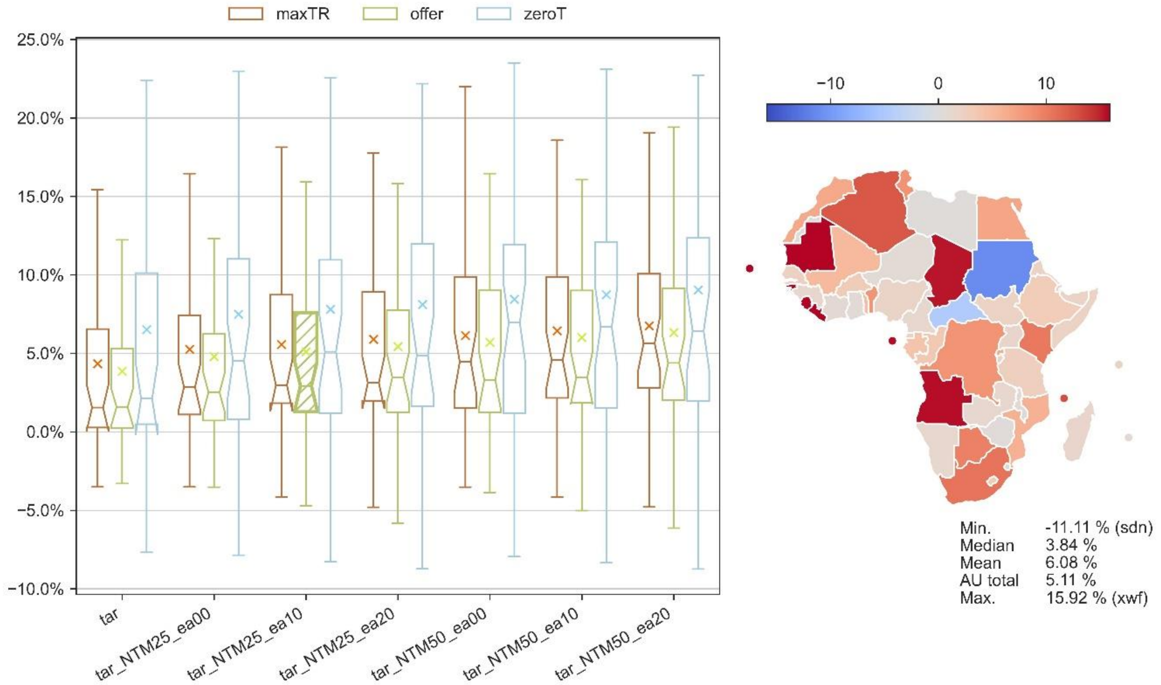
Figure 41. The AfCFTA effects on primary agricultural imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

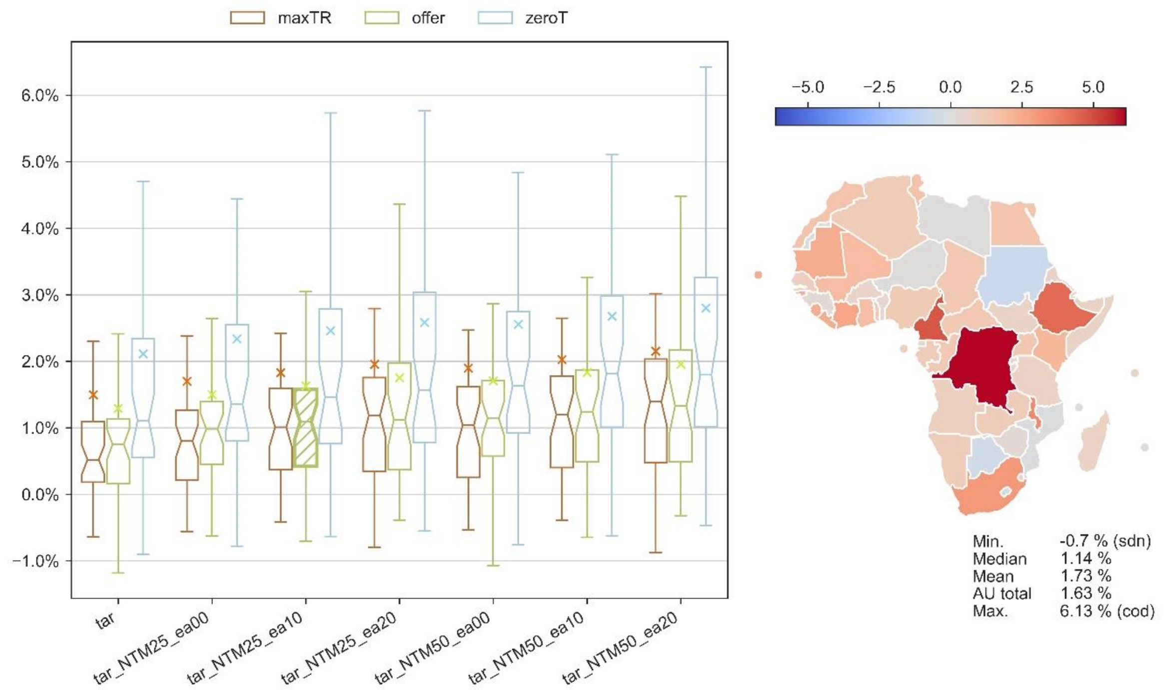
Both exports (Figure 42) and imports (Figure 43) of the processed food commodities increase on average, but with even more regional variation than with primary agricultural commodities. The highest increase in exports occurs in Rest of West Africa (16%) while the imports increase the most in Democratic Republic of Congo (6.1%).

Figure 42. The AfCFTA effects on processed food exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

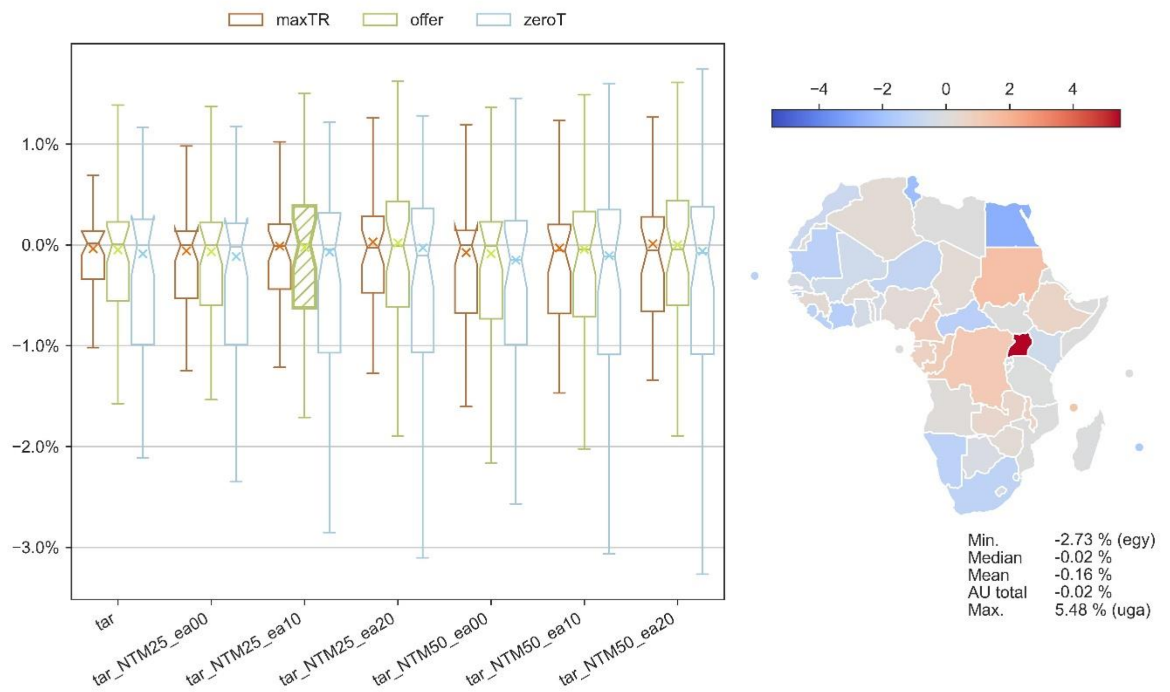
Figure 43. The AfCFTA effects on processed food imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

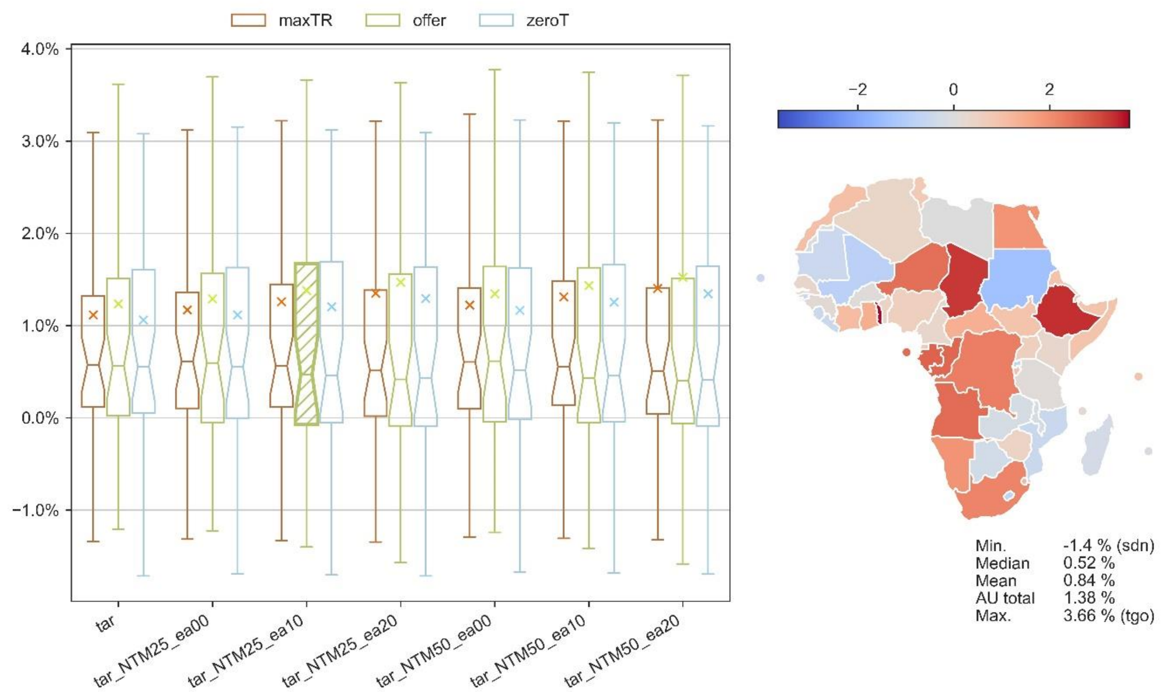
The non-renewable resource exports (Figure 44) hardly change on average, the mean change being slightly negative. The non-renewable resource imports (Figure 45), on the other hand, increase slightly. This is a consequence of increased incomes and demand for higher value-added commodities: due to the AfCFTA, Africa will use a higher share of its natural resources.

Figure 44. The AfCFTA effects on non-renewable resource exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

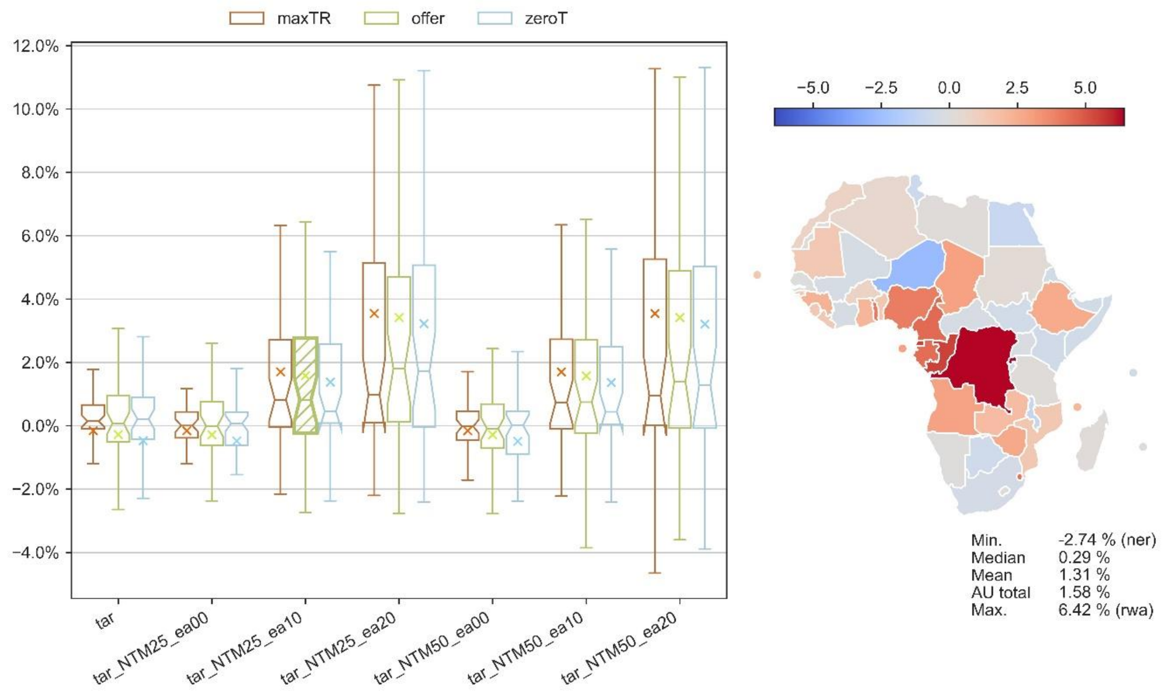
Figure 45. The AfCFTA effects on non-renewable resource imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

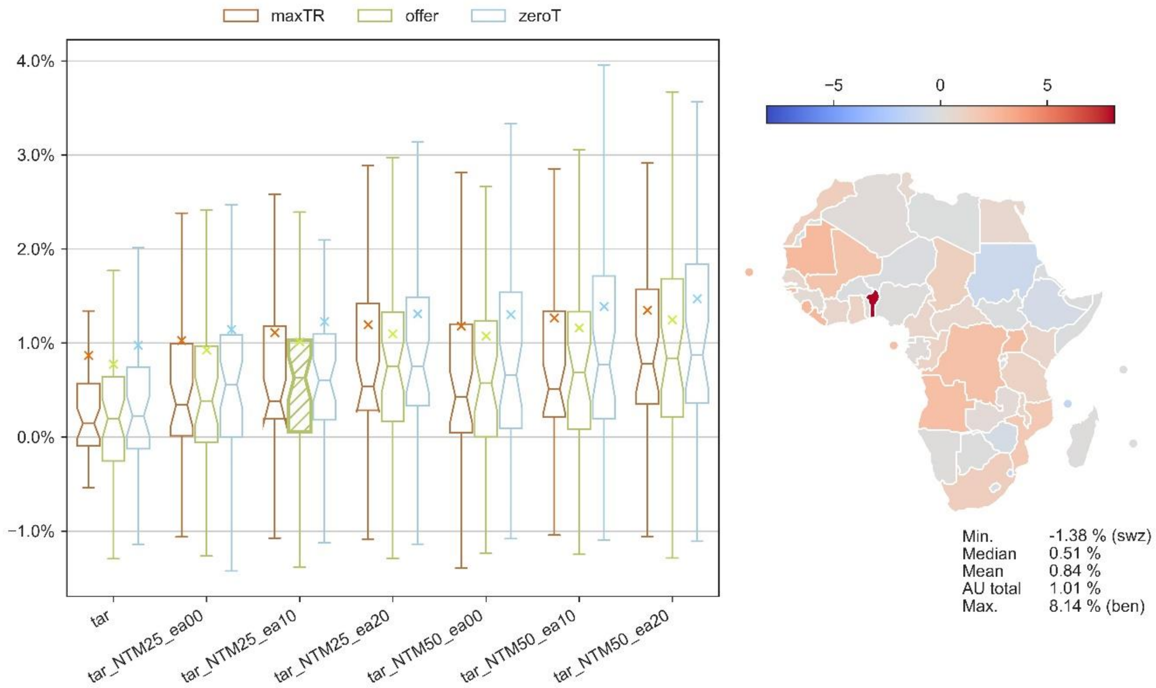
Both renewable resource exports (Figure 46) and imports (Figure 47) increase on average. Especially the exports are sensitive on the assumption of the spillover higher increases in exports being a result of higher spillovers. Regional variation is large, and the highest increase in exports happens in Rwanda (6.4%) and imports in Benin (8.1%).

Figure 46. The AfCFTA effects on renewable resource exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

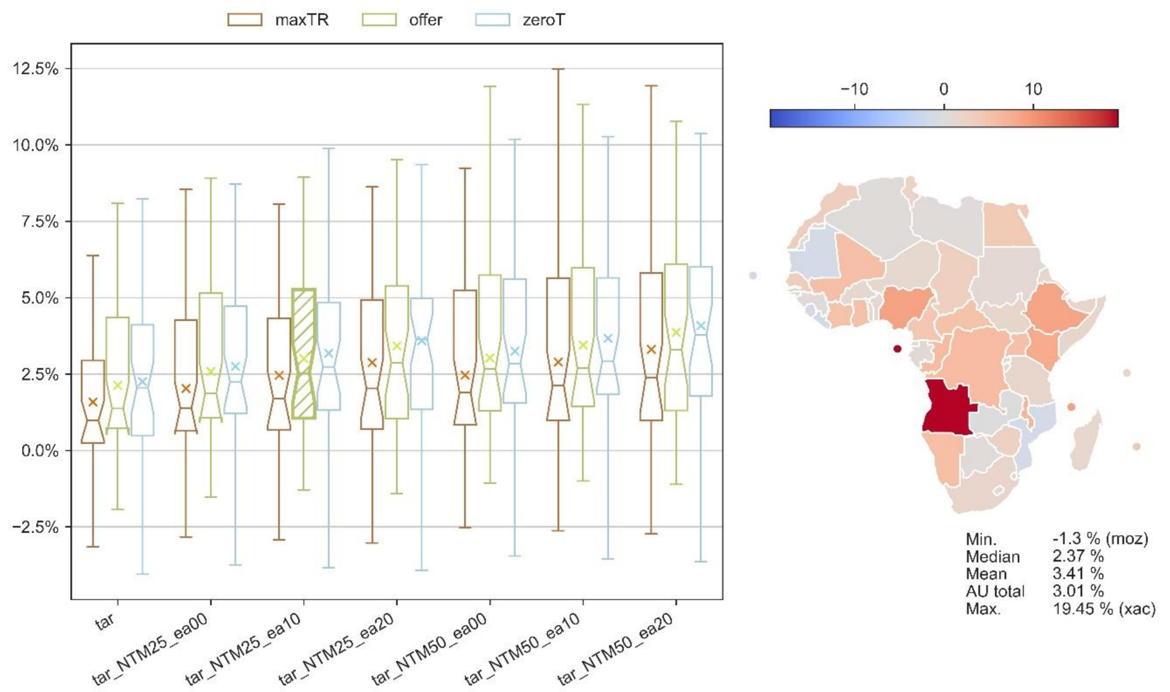
Figure 47. The AfCFTA effects on renewable resource imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

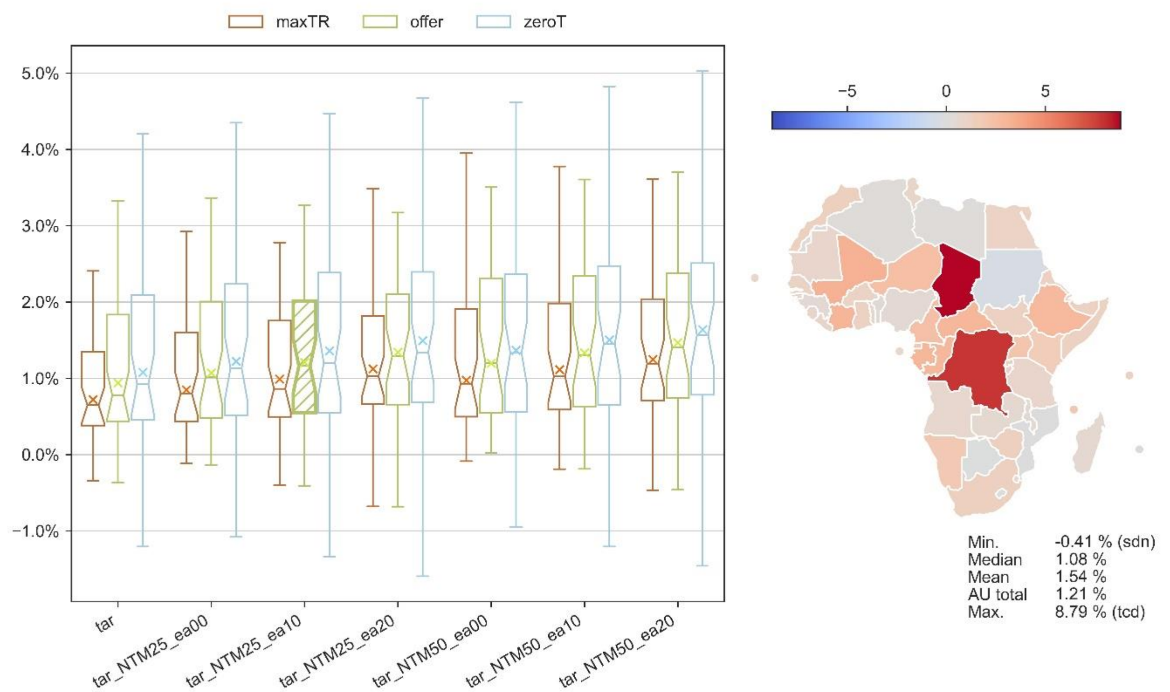
Both exports (Figure 48) and imports (Figure 49) of manufactures increase on average, although somewhat more with exports (the AU total 3.0%) than with imports (the AU total 1.2%). Therefore Africa will decrease its trade deficit in manufactured commodities due to the AfCFTA, another sign of the AfCFTA's boost on industrialization in Africa.

Figure 48. The AfCFTA effects on manufacturing exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

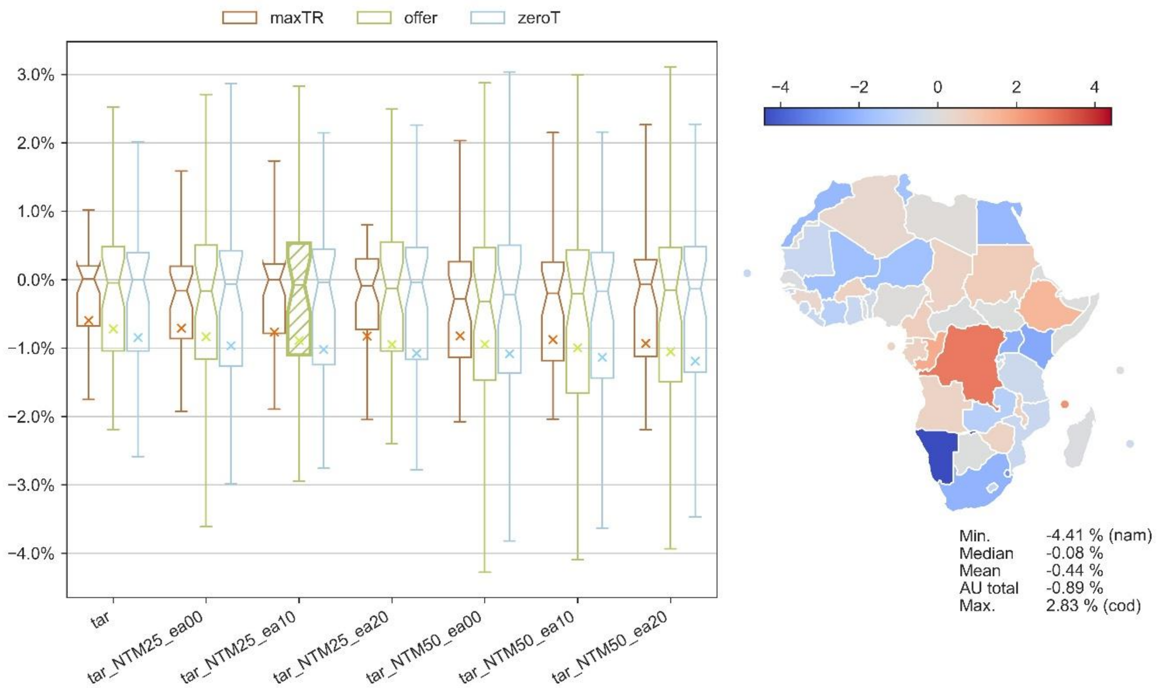
Figure 49. The AfCFTA effects on manufacturing imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

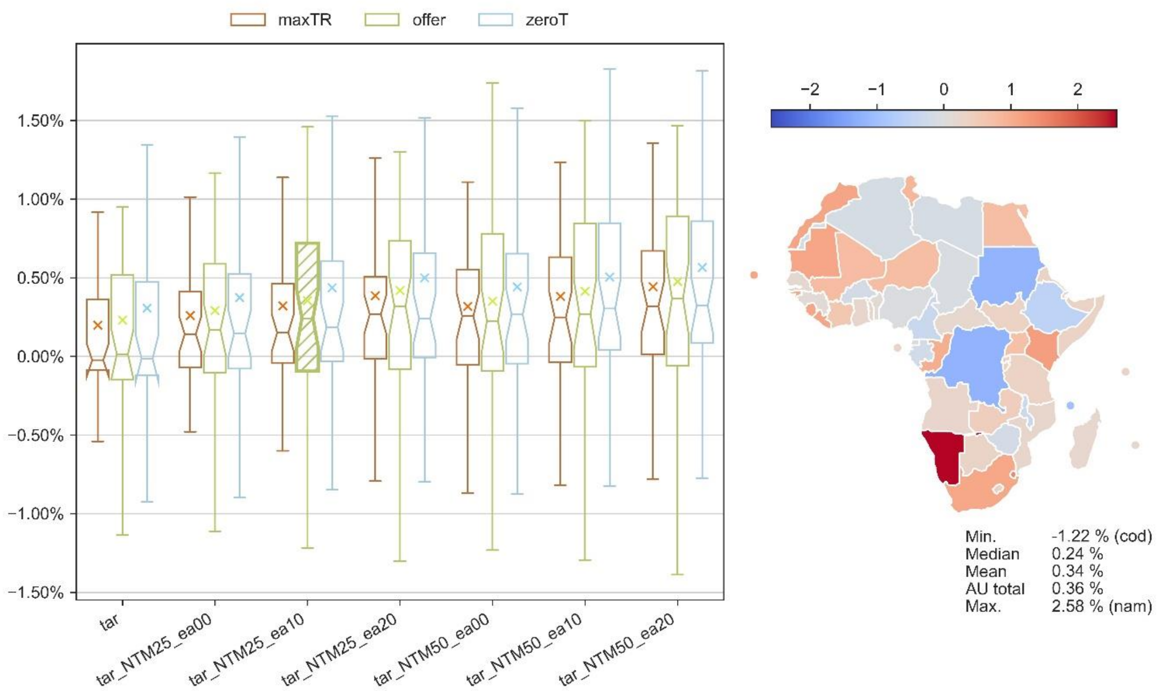
Finally, we can see that the exports (Figure 50) of the services decrease, while the imports (Figure 51) increase. The higher imports is a result of higher income levels, while the services sector needs to compete of the factors of production with other industries like manufacturing, and in the new equilibrium they will contract their exports.

Figure 50. The AfCFTA effects on services exports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Figure 51. The AfCFTA effects on services imports (percentage point deviation from baseline in 2035)



Source: the MAGNET calculations (note: the map shows the individual region results for our central scenario, which is hatched in the box plot; X marks the AU aggregate effect; outliers are omitted from the box plot).

Discussion

The results of the policy simulations provide a robust analysis of the potential impacts of the AfCFTA on Africa's economic landscape. The simulations suggest that the AfCFTA, once fully implemented, could serve as a catalyst for intra-African trade and economic integration, which is consistent with the agreement's objectives.

One of the key findings is the significant variation in the economic benefits across different regions and sectors. While the median GDP growth across the continent is modest, regions such as Rest of SACU and Congo are expected to experience more pronounced economic gains. This regional disparity underscores the importance of complementary policies that can ensure more equitable distribution of the benefits of trade liberalization.

The role of NTM reductions in enhancing the economic benefits of tariff reductions cannot be overstated. As shown in the results chapter, NTM reductions, especially when accompanied by spillover effects over extra-African trade, contribute to larger gains in household disposable income and GDP growth. This implies that for the AfCFTA to reach its full potential, there must be a concerted effort to address these non-tariff barriers alongside tariffs. Nevertheless, the analysis also suffers from the uncertainty surrounding the NTMs and their actionability, and the uncertainty on the potential spillover effect. Our results probably present an upper limit for what can be achieved with the NTM reductions. Small changes in the spillover effect have relatively large income effects. In fact, the spillovers affect the results more in relative terms than the NTM reductions alone. That indicates that by harmonizing NTMs with some non-African trade partners, additional increases in incomes could be achieved.[11] This is not a surprising outcome if we consider the still low share of intra-African trade. On the other hand, intra-African trade is somewhat reduced by increased spillovers. In addition, the spillovers are beneficial for increasing the output of services, while their effect on industrialization is negligible.

The potential loss in tariff revenues raises concerns about fiscal implications for the involved countries. However, the simulations suggest that the increase in domestic tax income due to stimulated economic activity may mitigate this loss. Ensuring the smooth transition of revenue streams will be critical for maintaining public spending and investment in the post-liberalization era. Liberalizing the trade further can yield small gains in GDP, but at the same time also further undermine public sector income.

The sector-specific analysis indicates that manufacturing may experience significant changes in output. This positive effect supports the AfCFTA’s goal for advancing industrialization in Africa. However, the services are affected much less and therefore the AfCFTA falls short of promoting a move to more services-oriented economy. In both manufacturing and services there is a wide regional variation. Thus, the overall industrialization thrust notwithstanding, there are regions that could fall behind in their economic structural change. These cases call for strategic assessment of natural economic advantages and perhaps as a result more active industrial policy. The agri-food sector is also positively affected by the AfCFTA, and this has implications for food security and rural development. These are critical issues for many African nations. As such, the AfCFTA's impact on the agricultural sector warrants further exploration to ensure that trade liberalization benefits are aligned with food security objectives.

This analysis, therefore, while focused on the tangible benefits of the AfCFTA, should be viewed as a crucial step within Africa's larger economic integration trajectory. As the United Nations Economic Commission for Africa (UNECA, AU and AfDB, 2025) articulates, the AfCFTA is designed as a foundational anchor for establishing an African Continental Customs Union (AfCCU) and an African

[11] Another interpretation is to consider the uncertainty and speculative nature of the spillovers: if such spillovers would fail to materialize along the AfCFTA, studies that assume such spillovers might artificially inflate the effects.

Continental Common Market (AfCCOM), moving towards a comprehensive African Economic Community. This broader, multi-stage vision for continental integration promises deeper economic ties and potentially more profound transformative impacts for Africa in the global arena. Further research into these subsequent phases of integration, building on the initial successes and lessons learned from the AfCFTA, will be essential for fully understanding and maximizing Africa's economic potential.

Conclusions

This study presents the most comprehensive and up-to-date analysis of the AfCFTA. The simulations conducted in the study indicate positive outcomes in terms of increased GDP, household disposable income, and intra-African trade. The AfCFTA represents a significant step towards economic integration in Africa, with the potential to yield substantial economic benefits for the continent. However, the full realization of these benefits is contingent upon the effective reduction of NTMs, a well-managed transition in public revenue streams, and the implementation of supportive measures to address regional and sectoral disparities.

As Africa moves towards greater economic integration, policymakers must ensure that the AfCFTA's implementation is accompanied by strategies that facilitate equitable growth, enhance competitiveness, and foster sustainable development. Future research should focus on the long-term social and environmental impacts of the AfCFTA, including its implications for employment, migration, regional disparities, and climate resilience[12].

This report also does not discuss the implementation and enforcement mechanisms of the AfCFTA. It needs to be kept in mind that we assume full implementation and enforcement of the agreement. A more careful analysis of the institutional and regulatory frameworks that are required to support the agreement is of considerable importance, but it is also something that we leave for future research endeavours.

In summary, while the AfCFTA has the potential to be a game-changer for African economies, its success will depend on the collective resolve of African nations to pursue a harmonized approach to trade liberalization, backed by policies that support structural transformation, capacity building, and inclusive growth.

[12] See Janssens et al. (2020 and 2022) for studies on this theme.

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Annexes

Annex 1: Regional aggregation

MAGNET region	Region name	GTAP region
dza	Algeria	dza
egy	Egypt	egy
mar	Morocco	mar
tun	Tunisia	tun
xnf	Rest of North Africa	xnf
ben	Benin	ben
bfa	Burkina Faso	bfa
cmr	Cameroon	cmr
civ	Cote d'Ivoire	civ
gha	Ghana	gha
gin	Guinea	gin
mli	Mali	mli
ner	Niger	ner
nga	Nigeria	nga
sen	Senegal	sen
tgo	Togo	tgo
xwf	Rest of Western Africa	xwf
caf	Central African Republic	caf
tcd	Chad	tcd
cog	Congo	cog
cod	Congo the Democratic Republic	cod

gnq	Equatorial Guinea	gnq
gab	Gabon	gab
xac	South Central Africa	xac
com	Comoros	com
eth	Ethiopia	eth
ken	Kenya	ken
mdg	Madagascar	mdg
mwi	Malawi	mwi
mus	Mauritius	mus
moz	Mozambique	moz
rwa	Rwanda	rwa
sdn	Sudan	sdn
tza	Tanzania	tza
uga	Uganda	uga
zmb	Zambia	zmb
zwe	Zimbabwe	zwe
xec	Rest of Eastern Africa	xec
bwa	Botswana	bwa
swz	Eswatini	swz
nam	Namibia	nam
zaf	South Africa	zaf
xsc	Rest of South African Customs Union	xsc

EU_27	EU27 regions	aut, bel, bgr, hrv, cyp, cze, dnk, est, fin, fra, deu, grc, hun, irl, ita, lva, ltu, lux, mlt, nld, pol, prt, rou, svk, svn, esp, swe
gbr	United Kingdom	gbr
RoEUR	Rest of Europe	che, nor,xef, alb, srb, xee, xer
FSU	Former Soviet Union	blr, rus, ukr, kaz, kgz, tjk, uzb, xsu, arm, geo
can	Canada	can
usa	United States of America	usa
OSA	Other S&C America	mex, xna, arg, bol, chl, col, ecu, pry, per, ury, ven, xsm, cri, gtm, hnd, nic, pan, slv, xca, dom, hti, jam, pri, tto, xcb
bra	Brazil	bra
chn	China	chn, hkg
SEA	South-East Asia	jpn, kor, twn, brn, khm, idn, lao, mys, phl, sgp, tha, vnm, xse
OAS	Other Asia	mng, xea, afg, bgd, npl, pak, lka, xsa
ind	India	ind
MET	Middle East	bhr, irn, irq, isr, jor, kwt, lbn, omn, pse, qat, sau, syr, tur, are, xws
ANZ	Australia / New Zealand	aus, nzl
ROW	Rest of the world	xoc, xtw

Annex 2: Sectoral/commodity aggregation

MAGNET com	Commodity name	GTAP commodity	Commodity group
pdr	Paddy rice	pdr	Primary agriculture
wht	Wheat	wht	Primary agriculture
gro	Cereal grains nec	gro	Primary agriculture
v_f	Vegetables, fruit, nuts	v_f	Primary agriculture
osd	Oil seeds	osd	Primary agriculture
c_b	Sugar cane, sugar beet	c_b	Primary agriculture
pfb	Plant-based fibers	pfb	Primary agriculture
ocr	Crops nec	ocr	Primary agriculture
ctl	Bovine cattle, sheep and goats	ctl	Primary agriculture
oap	Animal products nec	oap	Primary agriculture
rmk	Raw milk	rmk	Primary agriculture
wol	Wool, silk-worm cocoons	wol	Primary agriculture
frs	Forestry	frs	Renewable resources
fsh	Fishing	fsh	Renewable resources
coa	Coal	coa	Non-renewable resources
oil	Oil	oil	Non-renewable resources
gas	Gas	gas	Non-renewable resources
oxt	Minerals nec	oxt	Non-renewable resources
cmt	Bovine meat products	cmt	Processed food
omt	Meat products nec	omt	Processed food
vol	Vegetable oils and fats	vol	Processed food
mil	Dairy products	mil	Processed food
pcr	Processed rice	pcr	Processed food
sgr	Sugar	sgr	Processed food
ofd	Food products nec	ofd	Processed food
b_t	Beverages and tobacco products	b_t	Processed food
feed	Animal feed	ofd	Processed food
LightManuf	Light manufacturing	tex, wap, lea	Manufacturing
lum	Wood products	lum	Manufacturing
ppp	Paper products, publishing	ppp	Manufacturing
p_c	Petroleum, coal products	p_c	Manufacturing
chm	Chemical products	chm	Manufacturing
fert	Fertilizer	chm	Manufacturing
Manuf	Other manufacturing	bph, rpp, nmm, i_s, nfm, fmp, ele, eeq, ome, mvh, otn, omf	Manufacturing
ely	Electricity	ely	Utilities
gas_dist	Gas manufacture, distribution	gdt	Utilities
trd	Trade	trd	Services
foodserv	Accommodation and food services	afs	Services
Trnsp	Transport nec	otp, wtp, atp	Services
serv	Other services	wtr, cns, whs, cmn, ofi, ins, rsa, obs, ros, osg, edu, hht, dwe	Services

Annex 3: Online resources

The dashboard of the study results is available on the public website "EC Data-Modelling platform of resource economics DataM". Links can also be accessed with the QR code listed in this annex.

Figure A1: DataM URL – QR code:

https://datam.jrc.ec.europa.eu

Source: JRC, 2025.

Interactive dashboard

Users may explore and analyse the data through an interactive dashboard placed in the "PANAP network" section of the website.

Figure A2: DataM URL – QR code:

https://datam.jrc.ec.europa.eu/datam/mashup/AFCFTA_FOLLOWUP/index.html

Source: JRC, 2025.

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